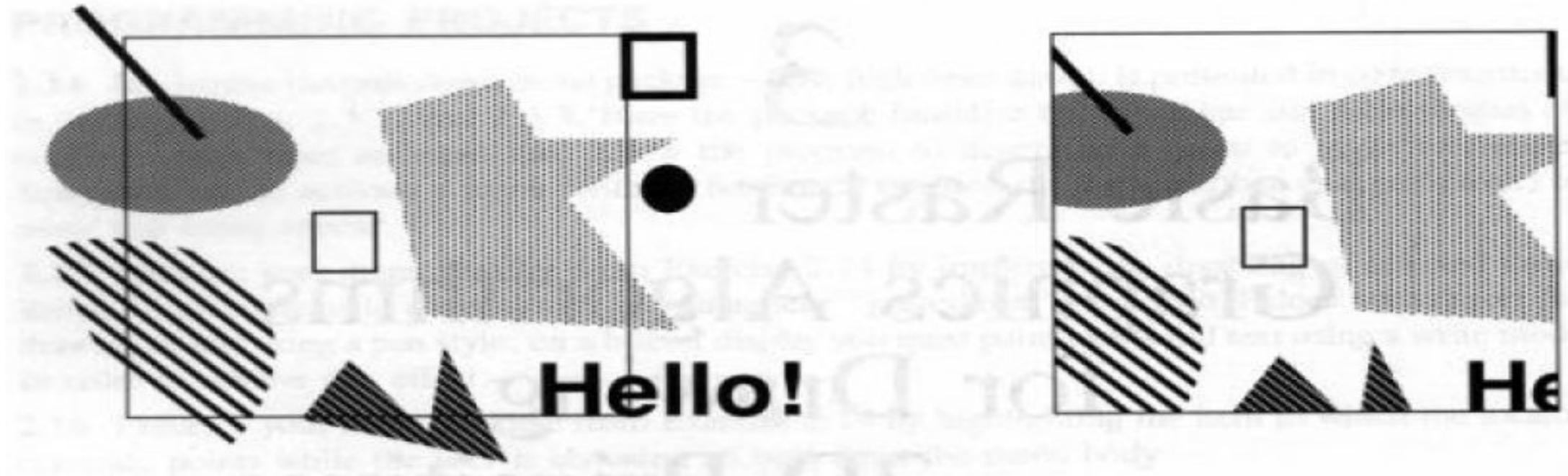


Line Clipping

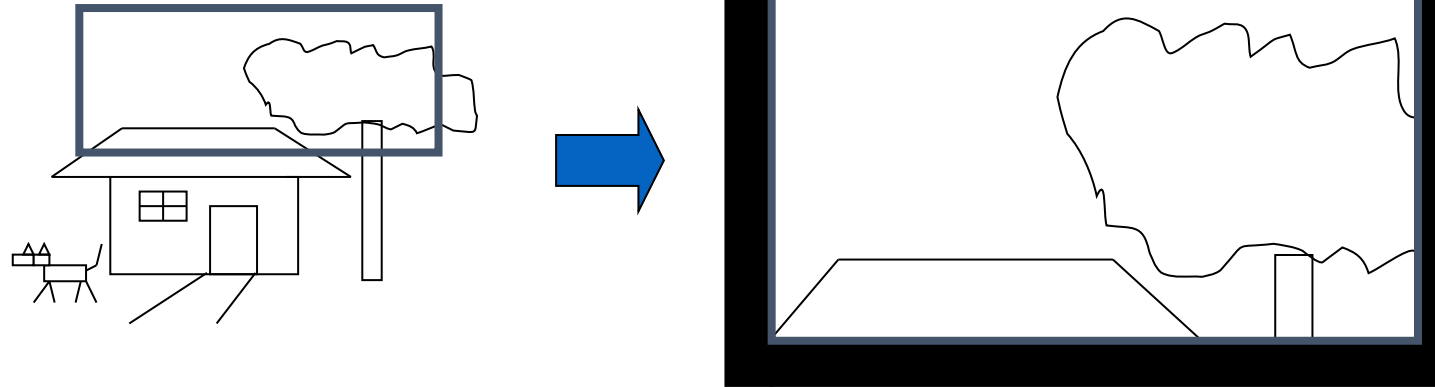
Cohen-Sutherland Algo



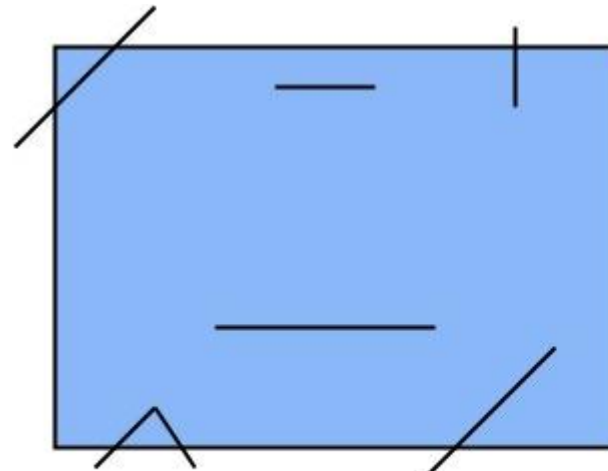
What is Clipping?



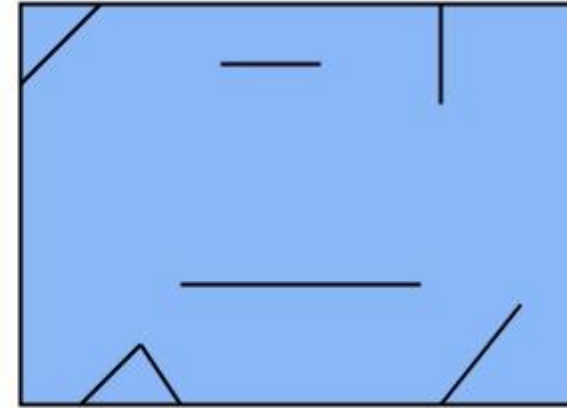
Clipped view in screen



Line Clipping



**Original Picture
or
Before Clipping**



After Clipping

Clipping in a Raster World

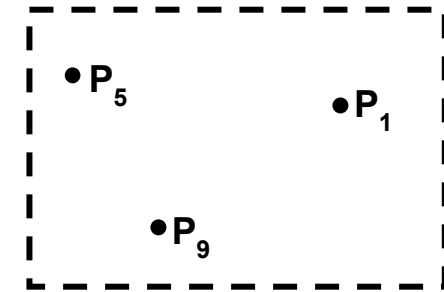
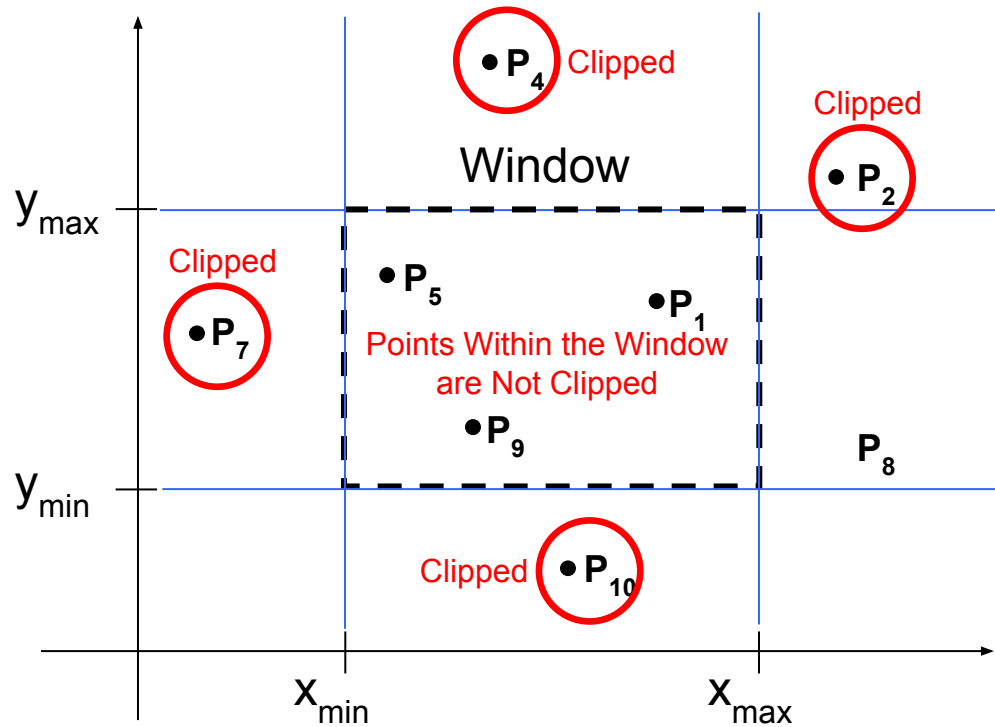
- Clipping techniques
 - **Analytical**
 - Lines, polygons etc.
 - Floating point graphics package
 - During scan conversion (scissoring)
 - Checking extrema suffices, internal points can be ignored
 - Circles, curves etc.
 - During writing a pixel
 - Outline primitive not much larger, few pixels are clipped

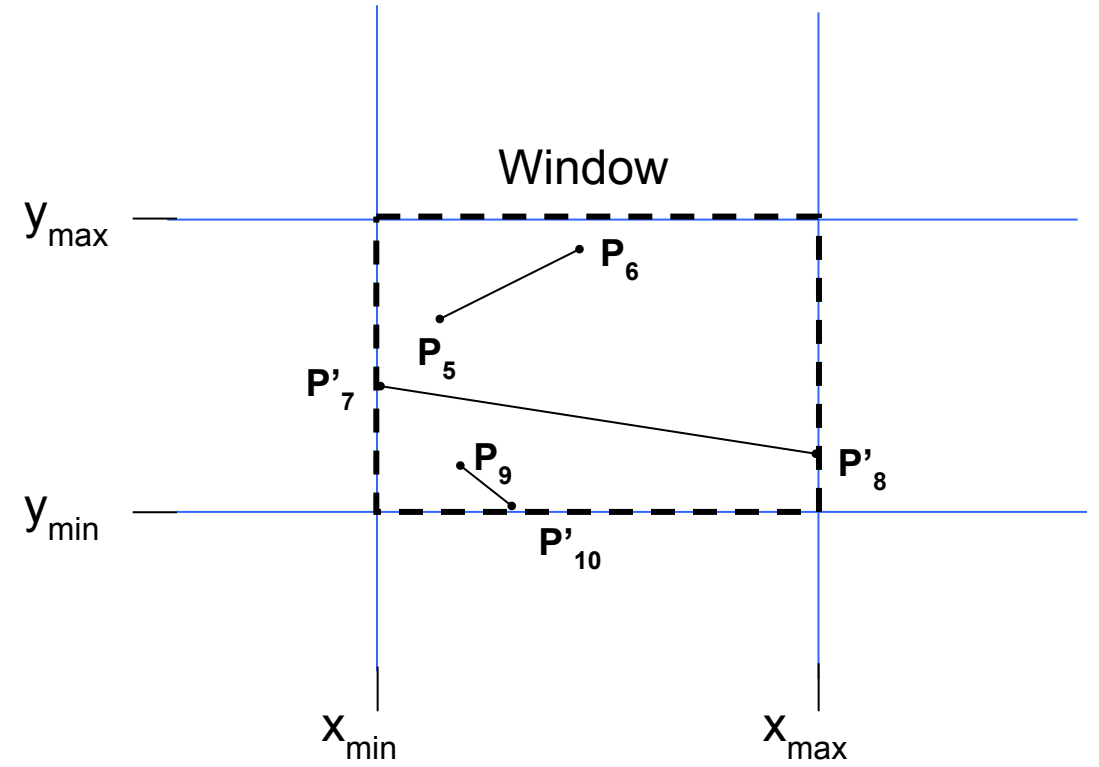
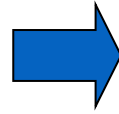
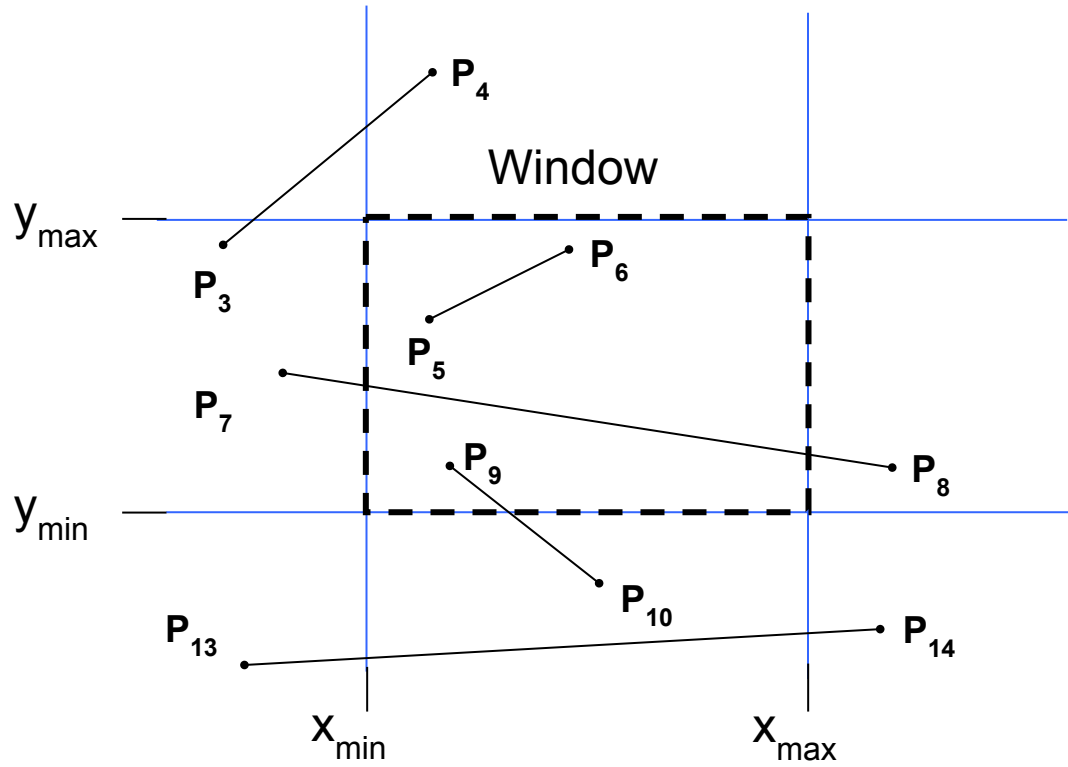


A point (x,y) is not clipped if:

$$x_{min} \leq x \leq x_{max} \text{ AND } y_{min} \leq y \leq y_{max}$$

otherwise it is clipped





Region Code/Outcode

World space is divided into regions based on the window boundaries

- Each region has a unique four bit outcode
- Outcodes indicate the position of the regions with respect to the window

3	2	1	0
above	below	right	left

Region Code

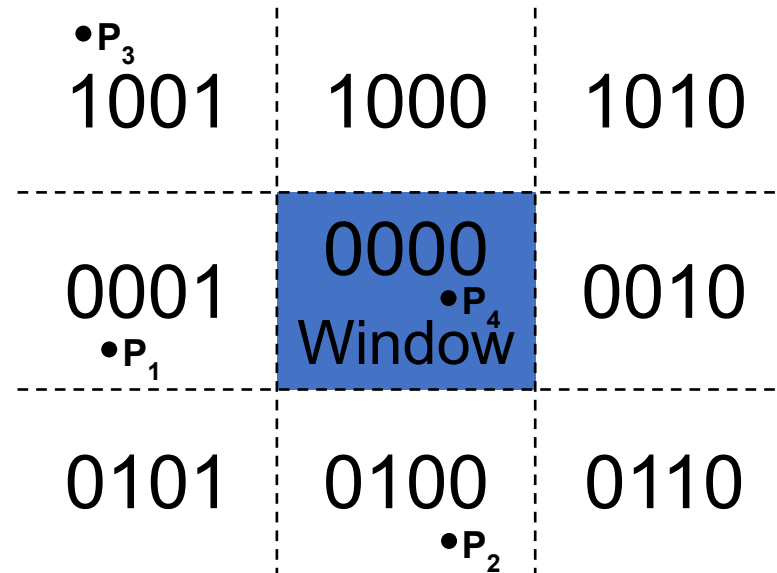
1001	1000	1010
0001	0000 Window	0010
0101	0100	0110

World space is divided into regions based on the window boundaries

- Each region has a unique four bit outcode
- Outcodes indicate the position of the regions with respect to the window

3	2	1	0
above	below	right	left

Region Code



Calculate_outcode(x,y){

if ($x < x_{\min}$) bit0 = 1

else bit0=0

if ($x > x_{\max}$) bit1=1

else bit1=0

if($y < y_{\min}$) bit2=1

else bit2=0

if ($y > y_{\max}$) bit3=1

else bit3=0

}

3	2	1	0
above	below	right	left

Region Code

Calculate_outcode(x,y){

if ($x < x_{\min}$) bit0 = 1

else bit0=0

if ($x > x_{\max}$) bit1=1

else bit1=0

if($y < y_{\min}$) bit2=1

else bit2=0

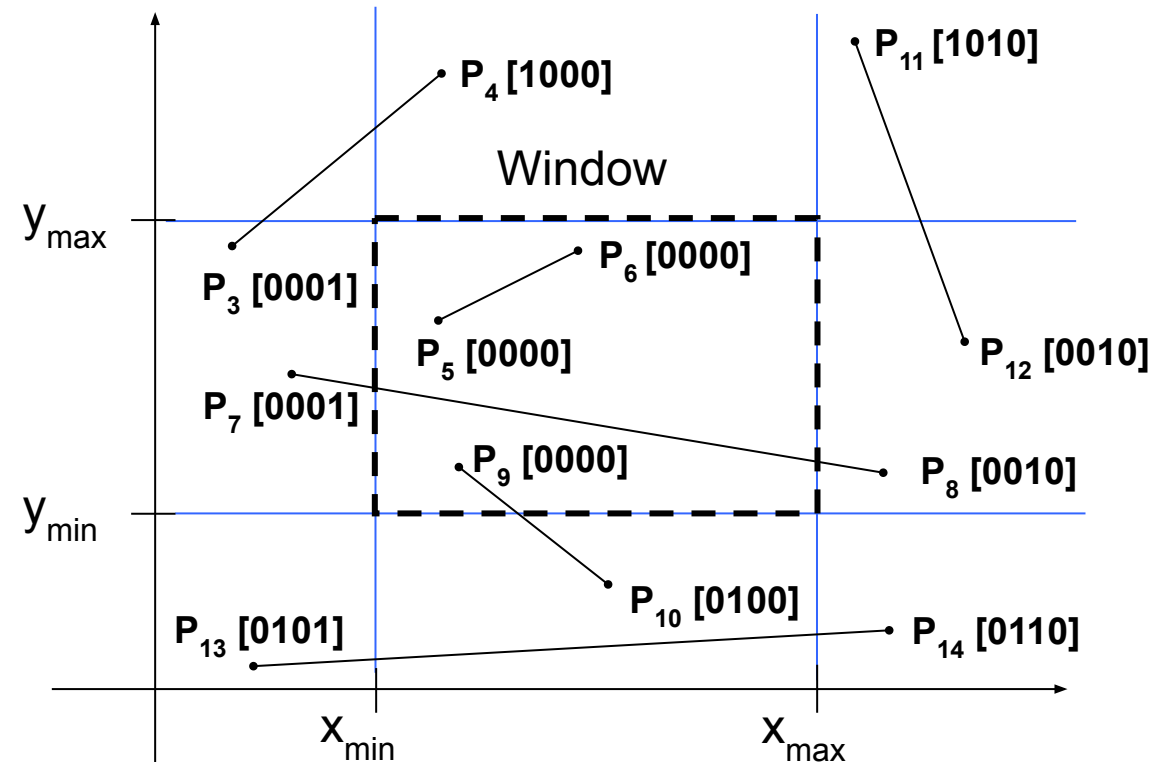
if ($y > y_{\max}$) bit3=1

else bit3=0

}

3	2	1	0
above	below	right	left

Region Code



```

Calculate_Outcode_3D(x,y,z){
    if (x<xmin) bit0 =1
    else bit0=0
    if (x>xmax) bit1=1
    else bit1=0
    if(y<ymin) bit2=1
    else bit2=0
    if (y>ymax) bit3=1
    else bit3=0
    if(z<zmin) bit4=1
    else bit4=0
    if (z>zmax) bit5=1
    else bit5=0
}

```

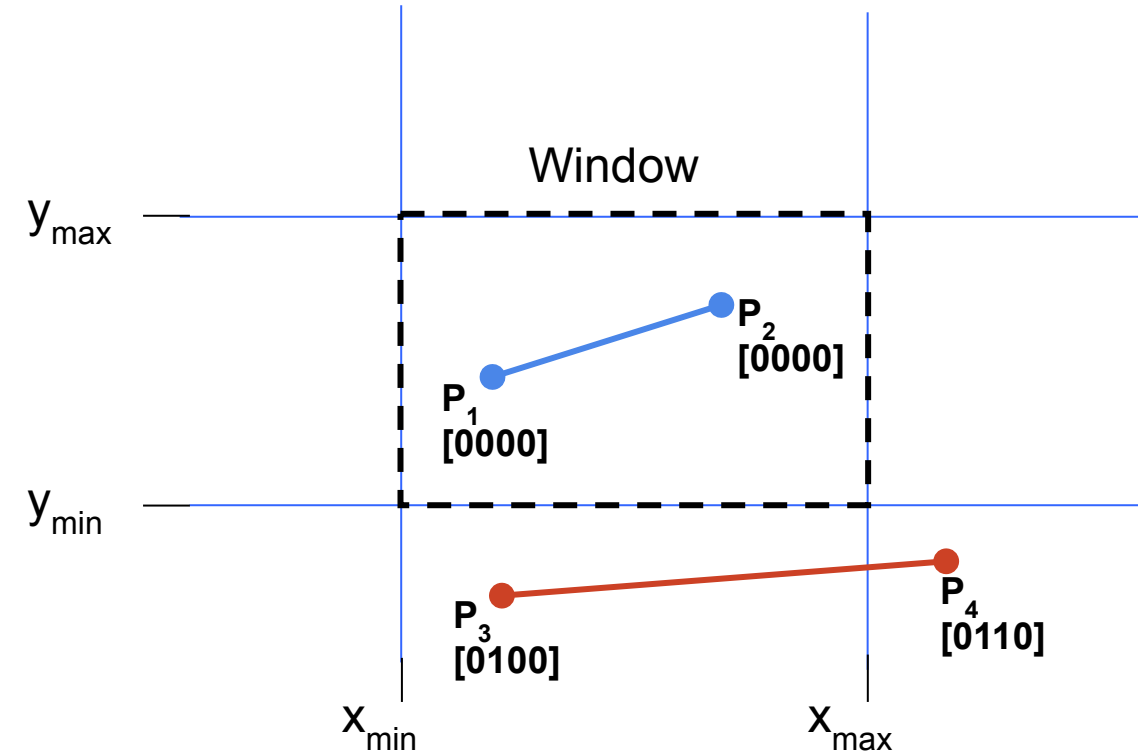
5	4	3	2	1	0
Near	Far	Above	Below	Right	Left
Region Code					

Condition of Trivial Acceptance:

- Lines that are completely **INSIDE**
- IF $OC1 == OC2 == 0000$

Condition of Trivial Rejection:

- Lines that are completely **OUTSIDE**
- IF $(OC1 \text{ AND } OC2) \neq 0000$



cohen-Sutherland(x1, y1, x2, y2):

oc1 = calculate_outcode(x1, y1),

oc2 = calculate_outcode(x2, y2);

while(true) {

 if(oc1 == oc2 == 0000) {

 //declare completely inside

 output (x1, y1), (x2, y2) as clipped line

 break

 }

 else if((oc1 AND oc2) != 0000){ // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if(oc1 != 0000){

 (x1, y1) = find intersection point of line

 and the boundary corresponding

 to non-zero bit of oc1

 oc1 = calculate_outcode(x1, y1)

 }

 else{

 (x2, y2) = find intersection point of line

 and the boundary corresponding

 to non-zero bit of oc2

 oc2 = calculate_outcode(x2, y2)

 }

 continue

}

cohen-Sutherland(x_1, y_1, x_2, y_2):

$oc1 = \text{calculate_outcode}(x_1, y_1)$,

$oc2 = \text{calculate_outcode}(x_2, y_2)$;

while(true) {

 if ($oc1 == oc2 == 0000$) {

 //declare completely inside

 output (x_1, y_1), (x_2, y_2) as clipped line

 break

 }

 else if ($(oc1 \text{ AND } oc2) \neq 0000$) { // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if($oc1 \neq 0000$){

 (x_1, y_1) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of $oc1$

$oc1 = \text{calculate_outcode}(x_1, y_1)$

 }

 else{

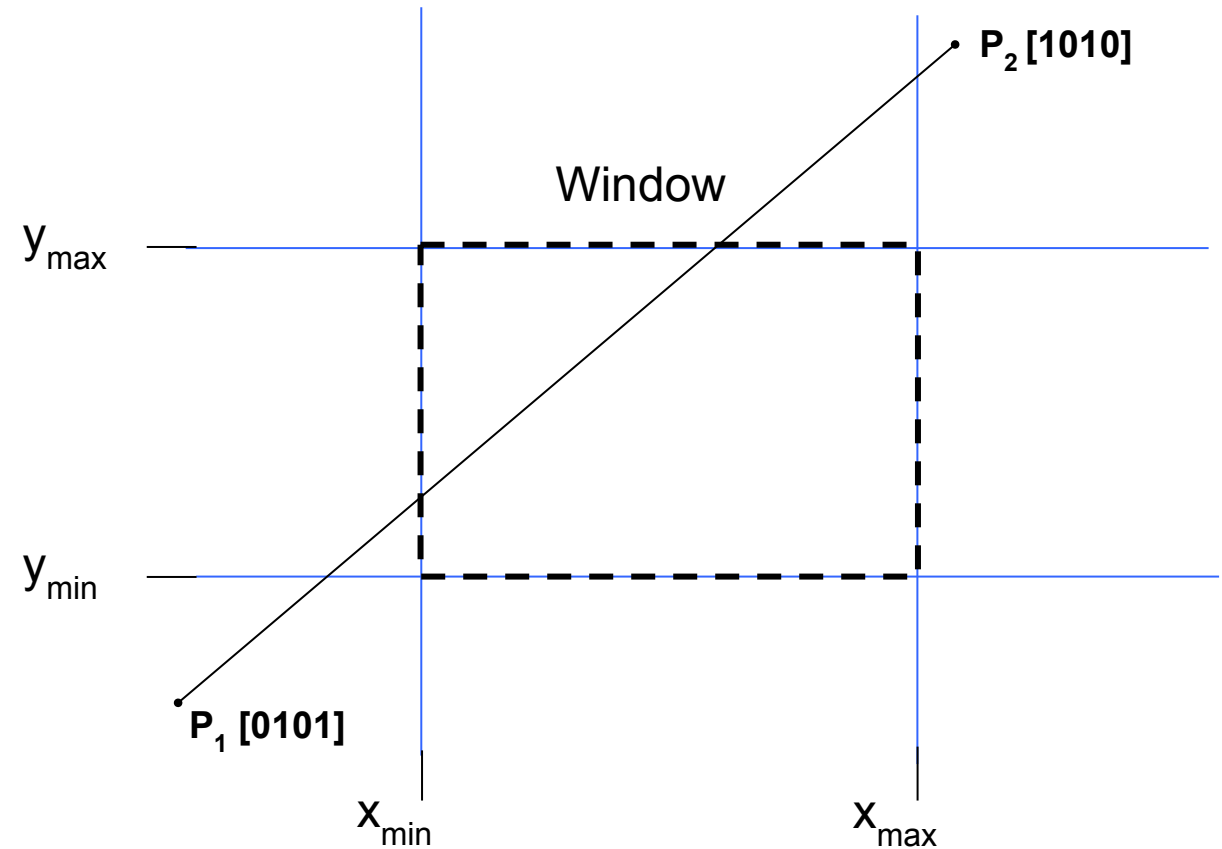
 (x_2, y_2) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of $oc2$

$oc2 = \text{calculate_outcode}(x_2, y_2)$

 }

 continue

}



cohen-Sutherland(x_1, y_1, x_2, y_2):

$oc1 = \text{calculate_outcode}(x_1, y_1)$,

$oc2 = \text{calculate_outcode}(x_2, y_2)$;

while(true) {

 if ($oc1 == oc2 == 0000$) {

 //declare completely inside

 output (x_1, y_1), (x_2, y_2) as clipped line

 break

 }

 else if ($(oc1 \text{ AND } oc2) \neq 0000$) { // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if($oc1 \neq 0000$){

 (x_1, y_1) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of $oc1$

$oc1 = \text{calculate_outcode}(x_1, y_1)$

 }

 else{

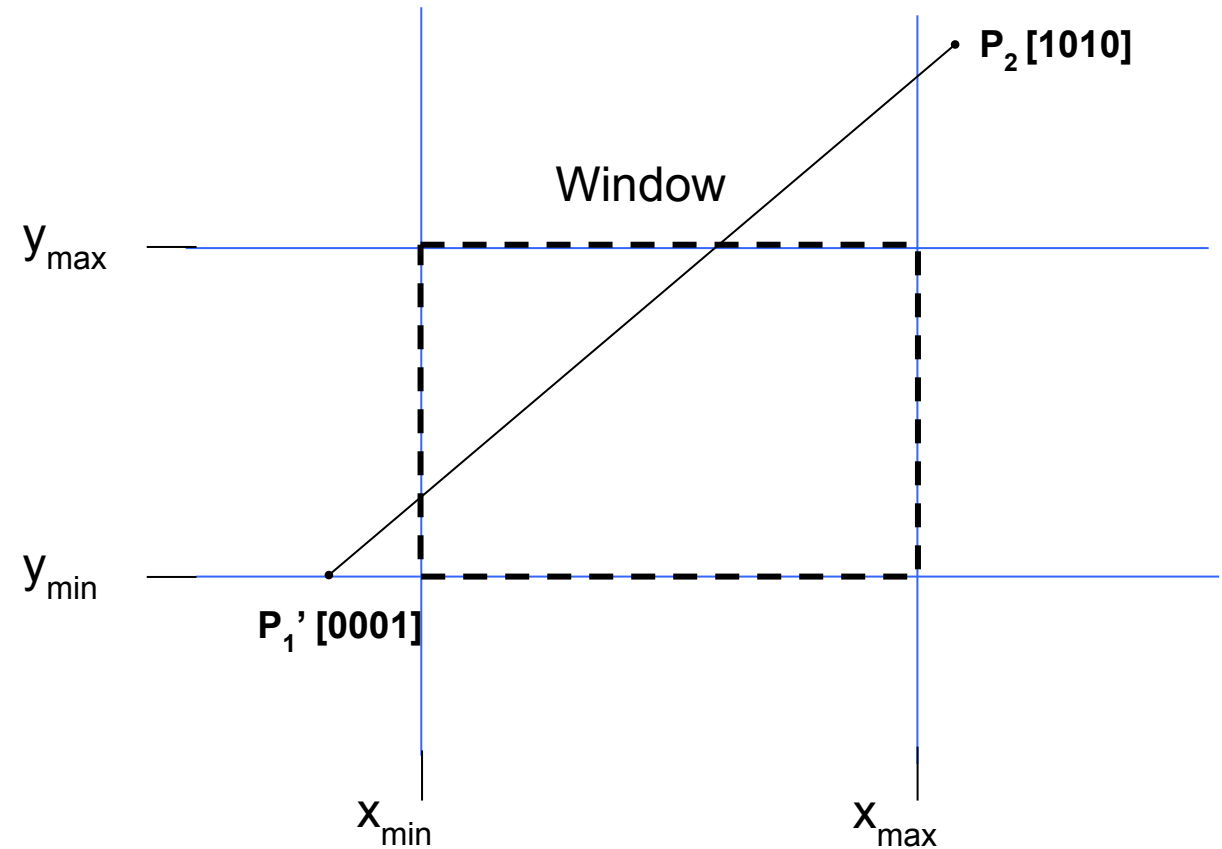
 (x_2, y_2) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of $oc2$

$oc2 = \text{calculate_outcode}(x_2, y_2)$

 }

 continue

}



cohen-Sutherland(x_1, y_1, x_2, y_2):

$oc_1 = \text{calculate_outcode}(x_1, y_1)$,

$oc_2 = \text{calculate_outcode}(x_2, y_2)$;

while(true) {

 if ($oc_1 == oc_2 == 0000$) {

 //declare completely inside

 output (x_1, y_1), (x_2, y_2) as clipped line

 break

 }

 else if ($(oc_1 \text{ AND } oc_2) \neq 0000$) { // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if($oc_1 \neq 0000$){

 (x_1, y_1) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of oc_1

$oc_1 = \text{calculate_outcode}(x_1, y_1)$

 }

 else{

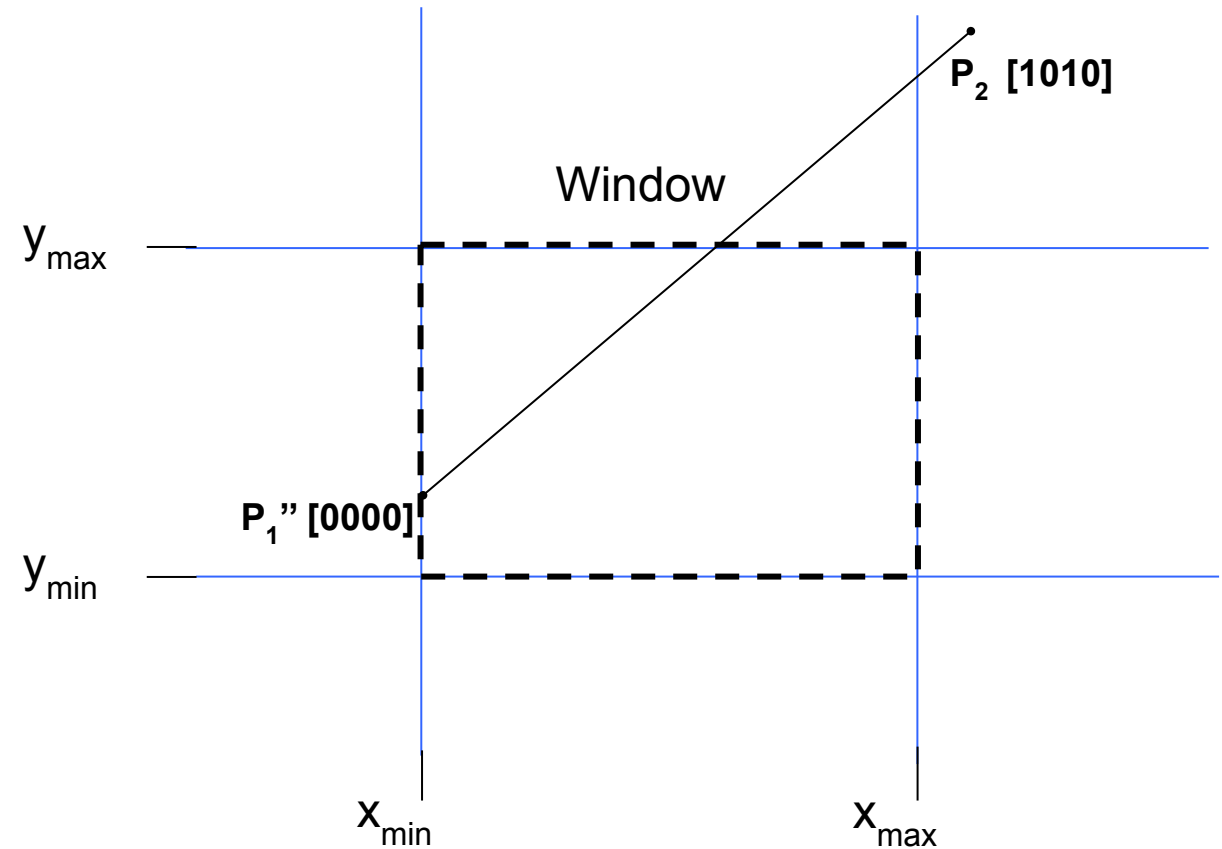
 (x_2, y_2) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of oc_2

$oc_2 = \text{calculate_outcode}(x_2, y_2)$

 }

 continue

}



cohen-Sutherland(x_1, y_1, x_2, y_2):

$oc_1 = \text{calculate_outcode}(x_1, y_1)$,

$oc_2 = \text{calculate_outcode}(x_2, y_2)$;

while(true) {

 if ($oc_1 == oc_2 == 0000$) {

 //declare completely inside

 output (x_1, y_1), (x_2, y_2) as clipped line

 break

 }

 else if ($(oc_1 \text{ AND } oc_2) \neq 0000$) { // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if($oc_1 \neq 0000$){

 (x_1, y_1) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of oc_1

$oc_1 = \text{calculate_outcode}(x_1, y_1)$

 }

 else{

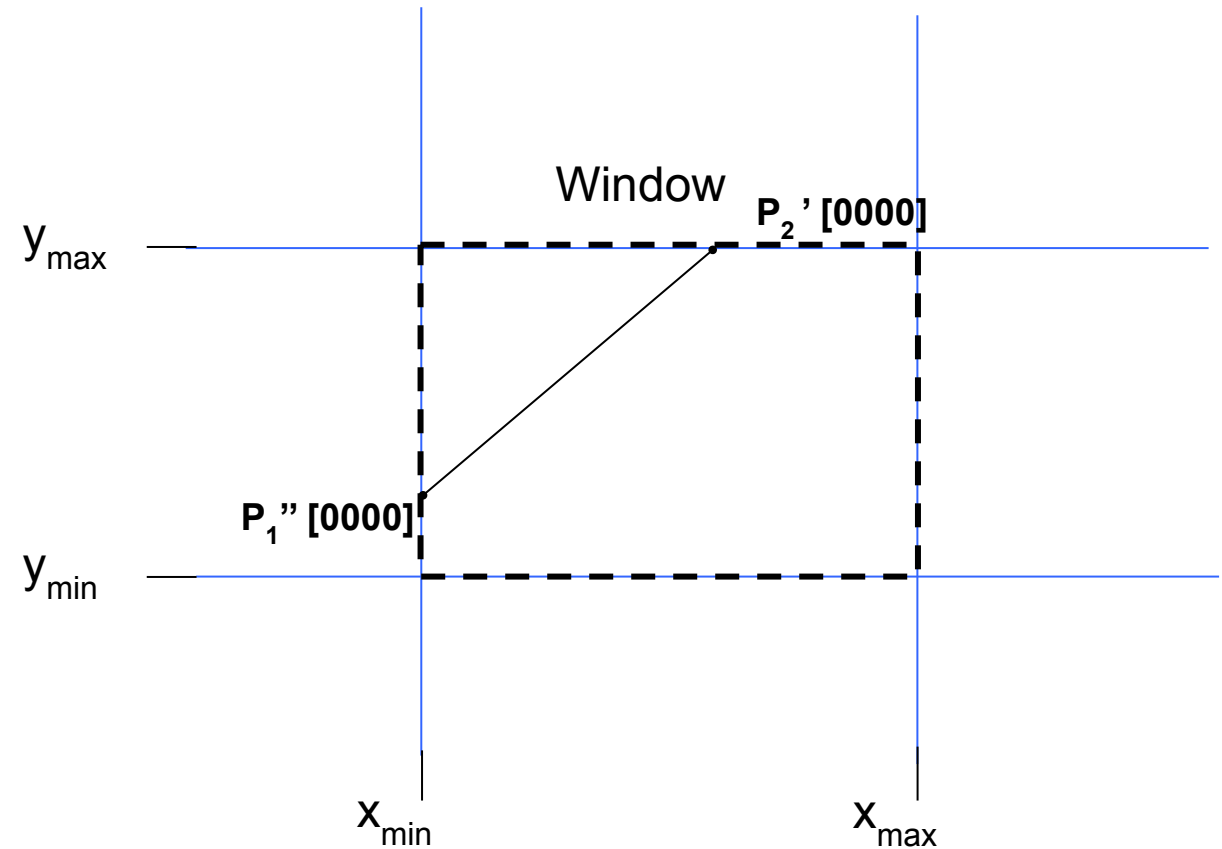
 (x_2, y_2) = find intersection point of line
 and the boundary corresponding
 to non-zero bit of oc_2

$oc_2 = \text{calculate_outcode}(x_2, y_2)$

 }

 continue

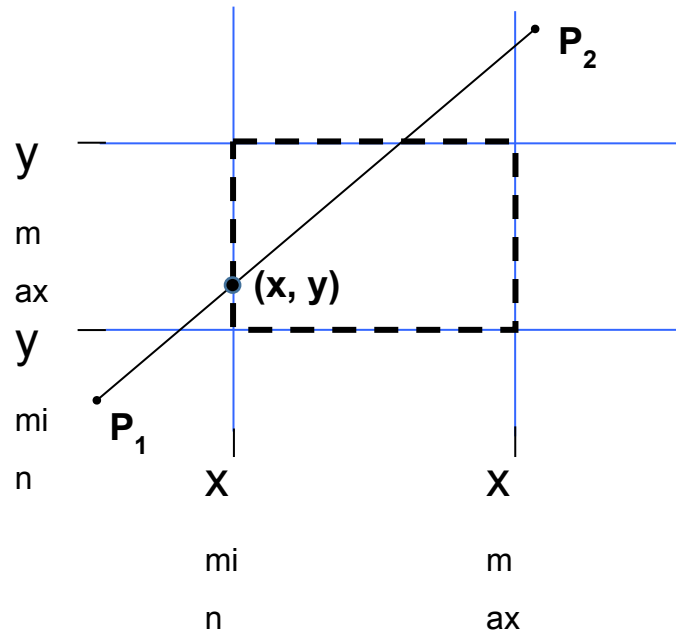
}



LEFT boundary intersection:

$$x = x_{\min}$$

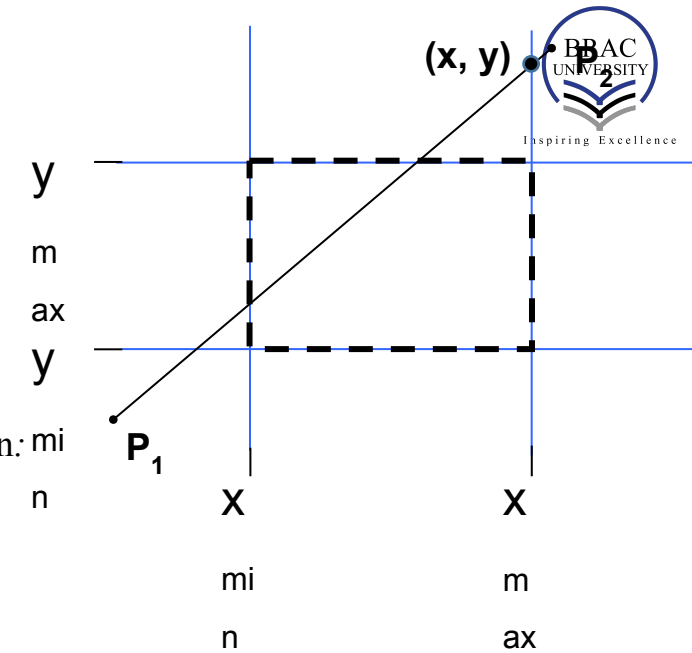
$$y = y_l + m (x_{\min} - x_l)$$



RIGHT boundary intersection:

$$x = x_{\max}$$

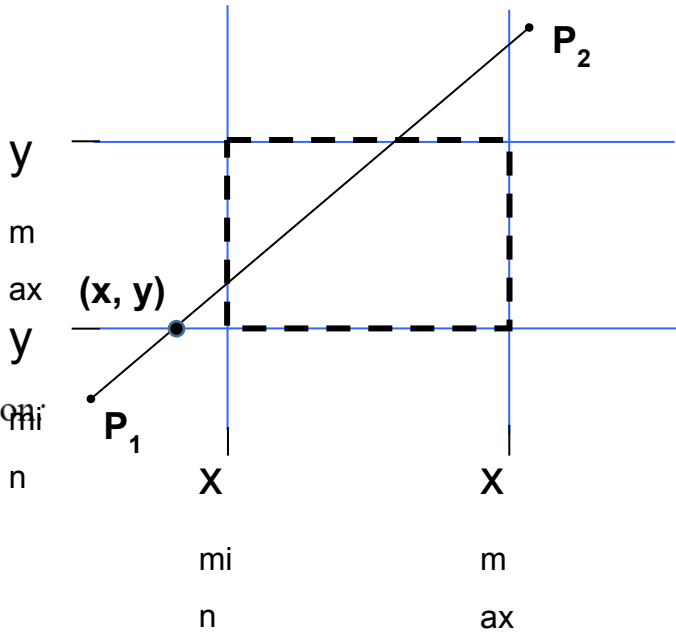
$$y = y_l + m (x_{\max} - x_l)$$



BOTTOM boundary intersection:

$$y = y_{\min}$$

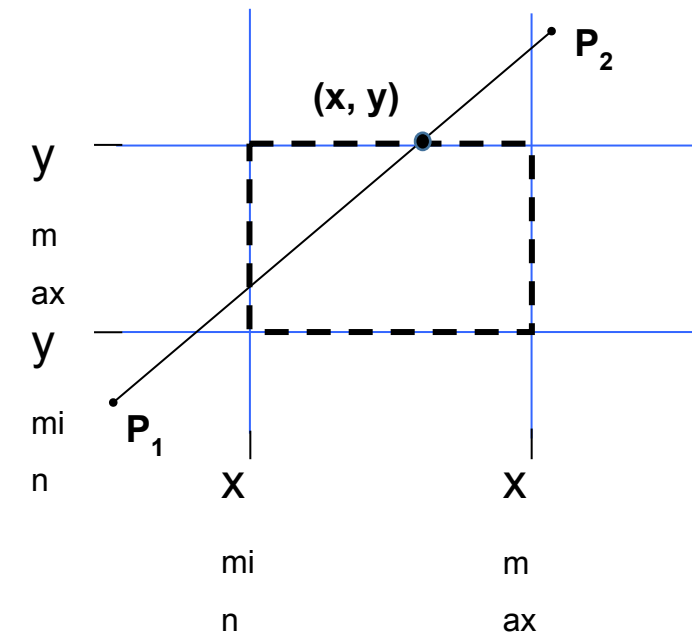
$$x = x_l + \frac{1}{m} \cdot (y_{\min} - y_l)$$



TOP boundary intersection:

$$y = y_{\max}$$

$$x = x_l + \frac{1}{m} \cdot (y_{\max} - y_l)$$



Determine whether the following line are accepted/rejected/partial using Cohen Sutherland line clipping algorithm.

a) Given $(-250, -200)$ to $(250, 200)$ be the clip region.

- (i) $(-100, -220)$ to $(300, -210)$.
- (ii) $(-250, 200)$ to $(250, -200)$.

b) Given $(0, 0)$ to $(300, 200)$ be the clip region.

- (i) $(50, -125)$ to $(-100, 225)$.
- (ii) $(-250, 200)$ to $(250, -200)$.

If they are partially accepted/rejected find the line segment within the clipping window.



a)(i) boundary:

$$x_{min} = -250, x_{max} = 250, y_{min} = -200, y_{max} = 200$$

points:

$$x_1 = -100, y_1 = -220, x_2 = 300, y_2 = -210$$

Outcode calculation:

$$x_{min} < x_1 < x_{max}$$

so, no left or right bit

$$y_1 < y_{min},$$

so, bottom bit is 1

so, outcode1 = 0100

$$x_1 > x_{max}$$

so, right bit

$$y_2 < y_{min}$$

so, bottom bit

so, outcode2 = 0110

$$\text{outcode1 AND outcode2} = 0100 \neq 0000$$

so the line is completely outside.

cohen-Sutherland(x_1, y_1, x_2, y_2):

oc1 = calculate_outcode(x_1, y_1),

oc2 = calculate_outcode(x_2, y_2);

```
while(true) {
    if (oc1 == oc2 == 0000) {
        //declare completely inside
        output (x1, y1), (x2, y2) as clipped line
        break
    }
    else if ((oc1 AND oc2) != 0000) { // condition to check matching bit
        //declare completely outside and clip
        break
    }
    else{
        if(oc1 != 0000){
            (x1, y1) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc1
            oc1 = calculate_outcode(x1, y1)
        }
        else{
            (x2, y2) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc2
            oc2 = calculate_outcode(x2, y2)
        }
        continue
    }
}
```

a)(ii) boundary:

$$x_{min} = -250, x_{max} = 250, y_{min} = -200, y_{max} = 200$$

points:

$$x_1 = -250, y_1 = 200, x_2 = 250, y_2 = -200$$

Outcode calculation:

$$x_{min} \leq x_1 < x_{max}$$

so, no left or right bit

$$y_{min} < y_1 \leq y_{max},$$

so, no top or bottom bit

so, outcode1 = 0000

similarly, outcode2 = 0000

since outcode1 & outcode 2 both are 0000

so the line is completely inside.

cohen-Sutherland(x_1, y_1, x_2, y_2):

oc1 = calculate_outcode(x_1, y_1),

oc2 = calculate_outcode(x_2, y_2);

```
while(true) {
    if (oc1 == oc2 == 0000) {
        //declare completely inside
        output (x1, y1), (x2, y2) as clipped line
        break
    }
    else if ((oc1 AND oc2) != 0000) { // condition to check matching bit
        //declare completely outside and clip
        break
    }
    else{
        if(oc1 != 0000){
            (x1, y1) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc1
            oc1 = calculate_outcode(x1, y1)
        }
        else{
            (x2, y2) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc2
            oc2 = calculate_outcode(x2, y2)
        }
        continue
    }
}
```

● b)(i) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_1 = 50, y_1 = -125, x_2 = -100, y_2 = 225$$

Outcode calculation:

$$\text{outcode1} = 0100$$

$$\text{outcode2} = 1001$$

$$\text{outcode1 AND outcode2} = 0000$$

so partially inside

$$\text{outcode1} \neq 0000$$

outcode1 has bottom bit

Applying bottom intersection:

$$y_1 = y_{min} = 0$$

$$x_1 = x_1 + \frac{1}{m} \cdot (y_{min} - y_1) = 50 + \frac{-150}{350} \cdot (0 + 125) = -3.57$$

$$\text{outcode1} = 0001 \text{ [recalculated]}$$

$$\text{outcode2} = 1001$$

$$\text{outcode1 AND outcode2} = 0001$$

so completely outside

cohen-Sutherland(x_1, y_1, x_2, y_2):

oc1 = calculate_outcode(x_1, y_1),

oc2 = calculate_outcode(x_2, y_2);

while(true) {

 if (oc1 == oc2 == 0000) {

 //declare completely inside

 output (x_1, y_1), (x_2, y_2) as clipped line

 break

 }

 else if ((oc1 AND oc2) != 0000) { // condition to check matching bit

 //declare completely outside and clip

 break

 }

 else{

 if(oc1 != 0000){

 (x_1, y_1) = find intersection point of line

 and the boundary corresponding

 to non-zero bit of oc1

 oc1 = calculate_outcode(x_1, y_1)

 }

 else{

 (x_2, y_2) = find intersection point of line

 and the boundary corresponding

 to non-zero bit of oc2

 oc2 = calculate_outcode(x_2, y_2)

 }

 continue

}

b)(ii) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_1 = -250, y_1 = 200, x_2 = 250, y_2 = -200$$

Outcode calculation:

outcode1 = 0001

outcode2 = 0100

outcode1 AND outcode2 = 0000

so partially inside

outcode1 != 0000

outcode1 has left bit

applying left intersection:

$$x_1 = x_{min} = 0$$

$$y_1 = y_1 + m \cdot (x_{min} - x_1) = 200 + \frac{-400}{500} \cdot (0 + 250) = 0$$

outcode1 = 0000 [recalculated]

so (x1, y1) has been clipped to (0, 0)

outcode2 = 0100

so partially inside

cohen-Sutherland(x1, y1, x2, y2):

oc1 = calculate_outcode(x1, y1),

oc2 = calculate_outcode(x2, y2);

```
while(true) {
    if (oc1 == oc2 == 0000) {
        //declare completely inside
        output (x1, y1), (x2, y2) as clipped line
        break
    }
    else if ((oc1 AND oc2) != 0000) { // condition to check matching bit
        //declare completely outside and clip
        break
    }
    else{
        if(oc1 != 0000){
            (x1, y1) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc1
            oc1 = calculate_outcode(x1, y1)
        }
        else{
            (x2, y2) = find intersection point of line
                        and the boundary corresponding
                        to non-zero bit of oc2
            oc2 = calculate_outcode(x2, y2)
        }
        continue
    }
}
```


b)(ii) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_1 = 0, y_1 = 0, x_2 = 250, y_2 = -200$$

(continued)

outcode1 = 0000 [recalculated]

outcode2 = 0100

so partially inside

outcode1 = 0000

so going into else codeblock,

outcode 2 has bottom bit

applying bottom intersection:

$$y_2 = y_{min} = 0$$

$$x_2 = x_2 + \frac{1}{m} \cdot (y_{min} - y_2) = 250 + \frac{250}{-200} \cdot (0 + 200) = 0$$

[Note: m has been recalculated, you can skip recalculation too]

outcode2 = 0000

outcode1=0000

so completely inside

The clipped segment is between (0, 0) to (0, 0) which is just a single point.

cohen-Sutherland(x1, y1, x2, y2):

oc1 = calculate_outcode(x1, y1),

oc2 = calculate_outcode(x2, y2);

while(true) {

if (oc1 == oc2 == 0000) {

//declare completely inside

output (x1, y1), (x2, y2) as clipped line

break

}

else if ((oc1 AND oc2) != 0000) { // condition to check matching bit

//declare completely outside and clip

break

}

else{

if(oc1 != 0000){

(x1, y1) = find intersection point of line
and the boundary corresponding
to non-zero bit of oc1

oc1 = calculate_outcode(x1, y1)

}

else{

(x2, y2) = find intersection point of line
and the boundary corresponding
to non-zero bit of oc2

oc2 = calculate_outcode(x2, y2)

}

continue

}

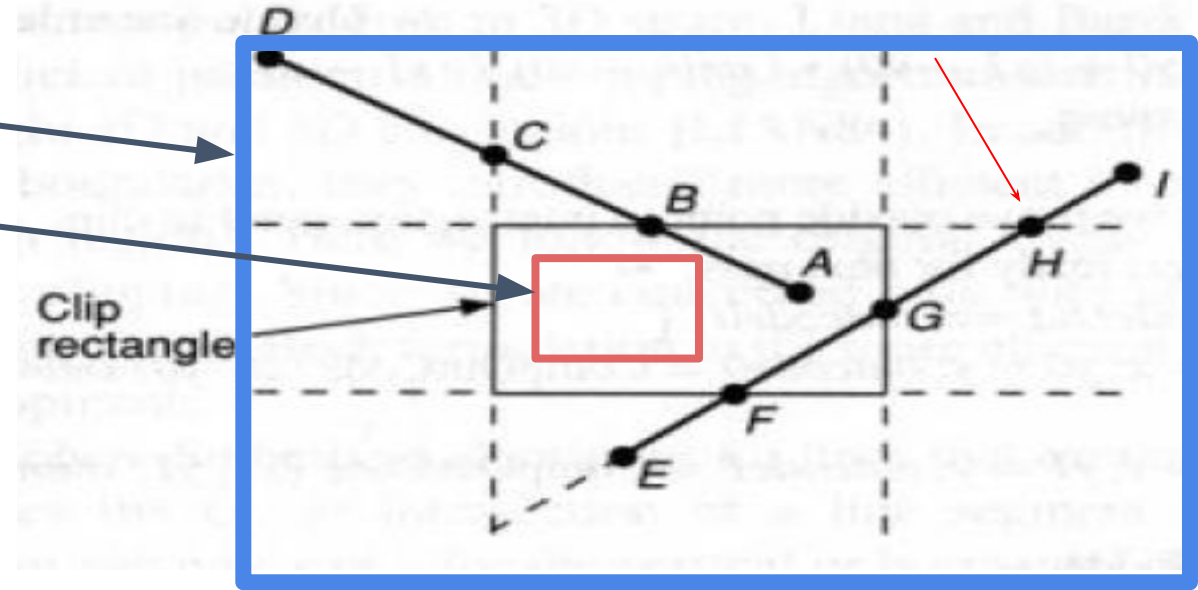
Cohen-Sutherland Line Clipping Algorithm

Works well for two cases:

1. Very large clip region
2. Very small clip region

- Why?

[For many trivial accept and many trivial reject]



Where is the problem?

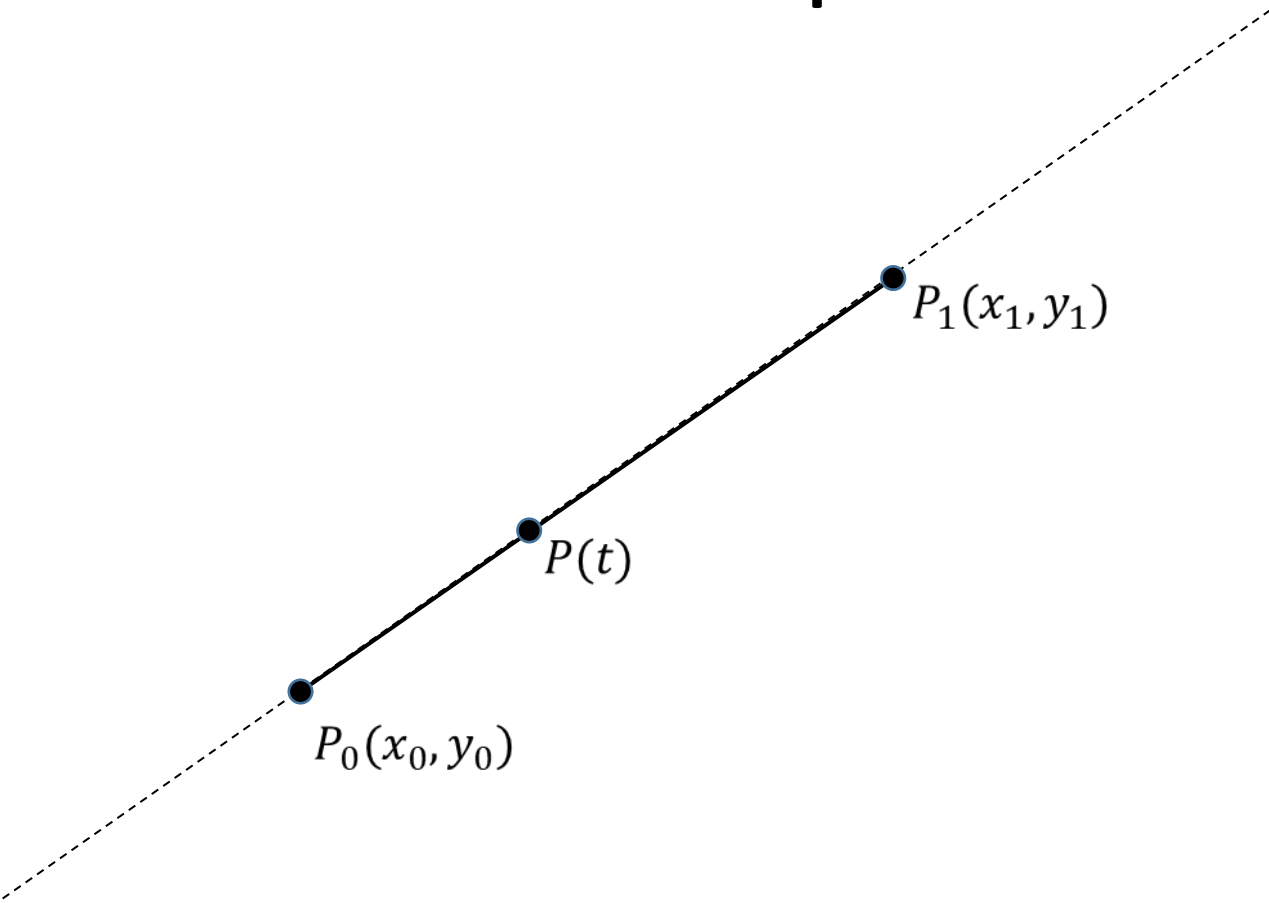
- Only rectangular clipping region
- Unnecessary clipping is done
- Different clipping order may take less iterations to finish

Line Clipping

Cyrus-Beck Algo



Parametric equation of Line



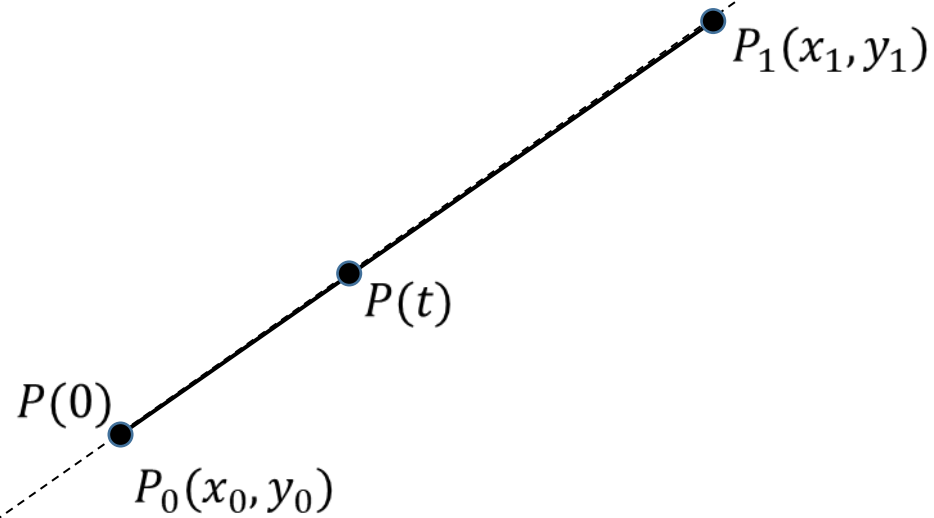
Parametric equation,

$$P(t) = P_0 + t \cdot (P_1 - P_0)$$

$$= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0)$$

$$= (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0))$$

Parametric equation of Line

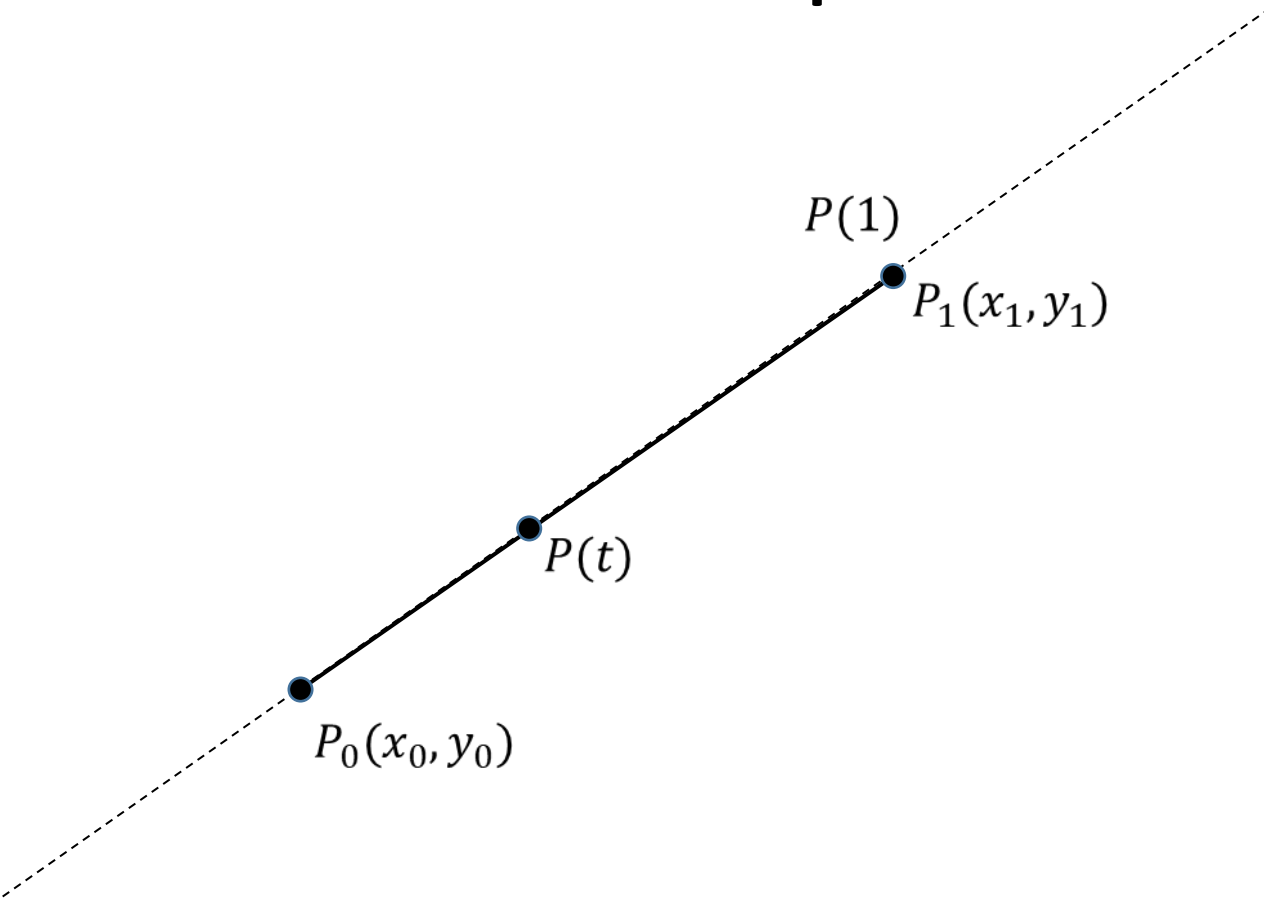


Parametric equation,

$$\begin{aligned} P(t) &= P_0 + t \cdot (P_1 - P_0) \\ &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\ &= (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0)) \end{aligned}$$

$$\begin{aligned} P(0) &= (x_0, y_0) + 0 \cdot (x_1 - x_0, y_1 - y_0) \\ &= (x_0, y_0) \end{aligned}$$

Parametric equation of Line

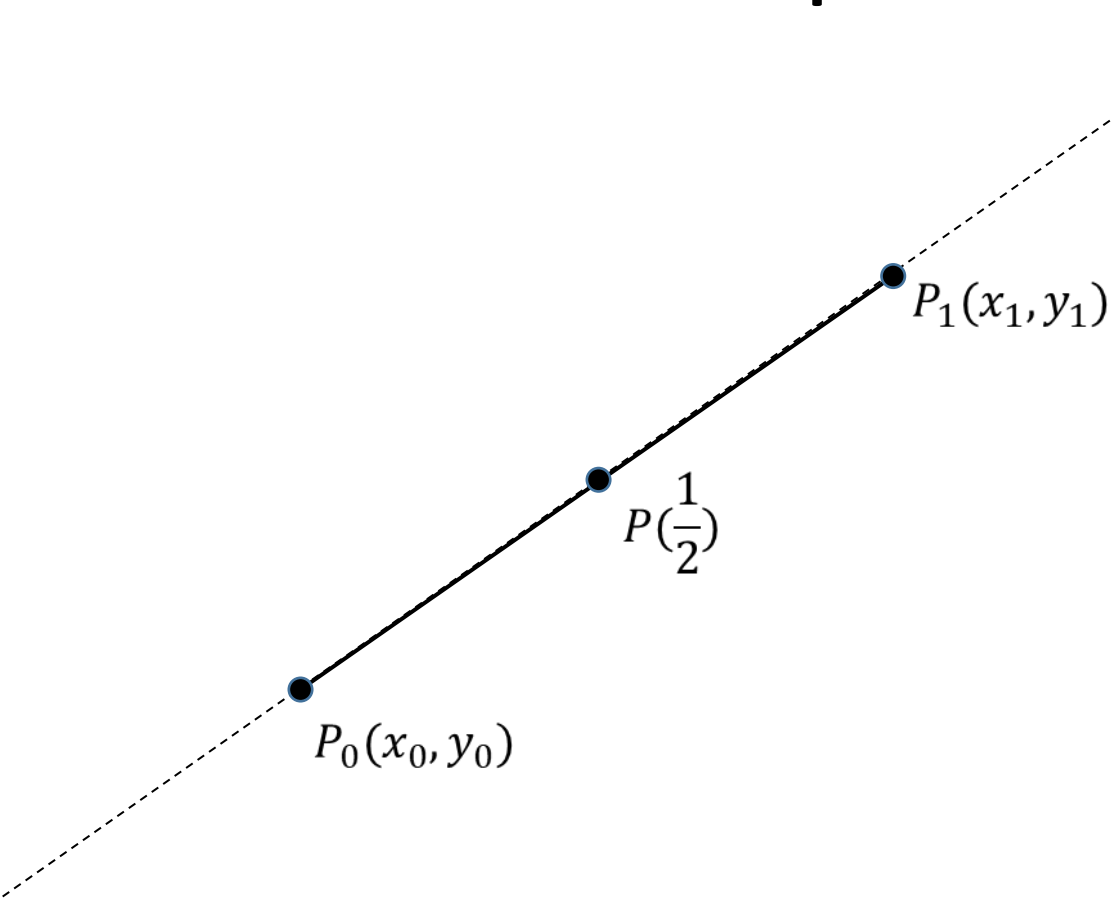


Parametric equation,

$$\begin{aligned} P(t) &= P_0 + t \cdot (P_1 - P_0) \\ &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\ &= (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0)) \end{aligned}$$

$$\begin{aligned} P(1) &= (x_0, y_0) + 1 \cdot (x_1 - x_0, y_1 - y_0) \\ &= (x_0, y_0) + (x_1 - x_0, y_1 - y_0) \\ &= (x_0 + x_1 - x_0, y_0 + y_1 - y_0) \\ &= (x_1, y_1) \end{aligned}$$

Parametric equation of Line

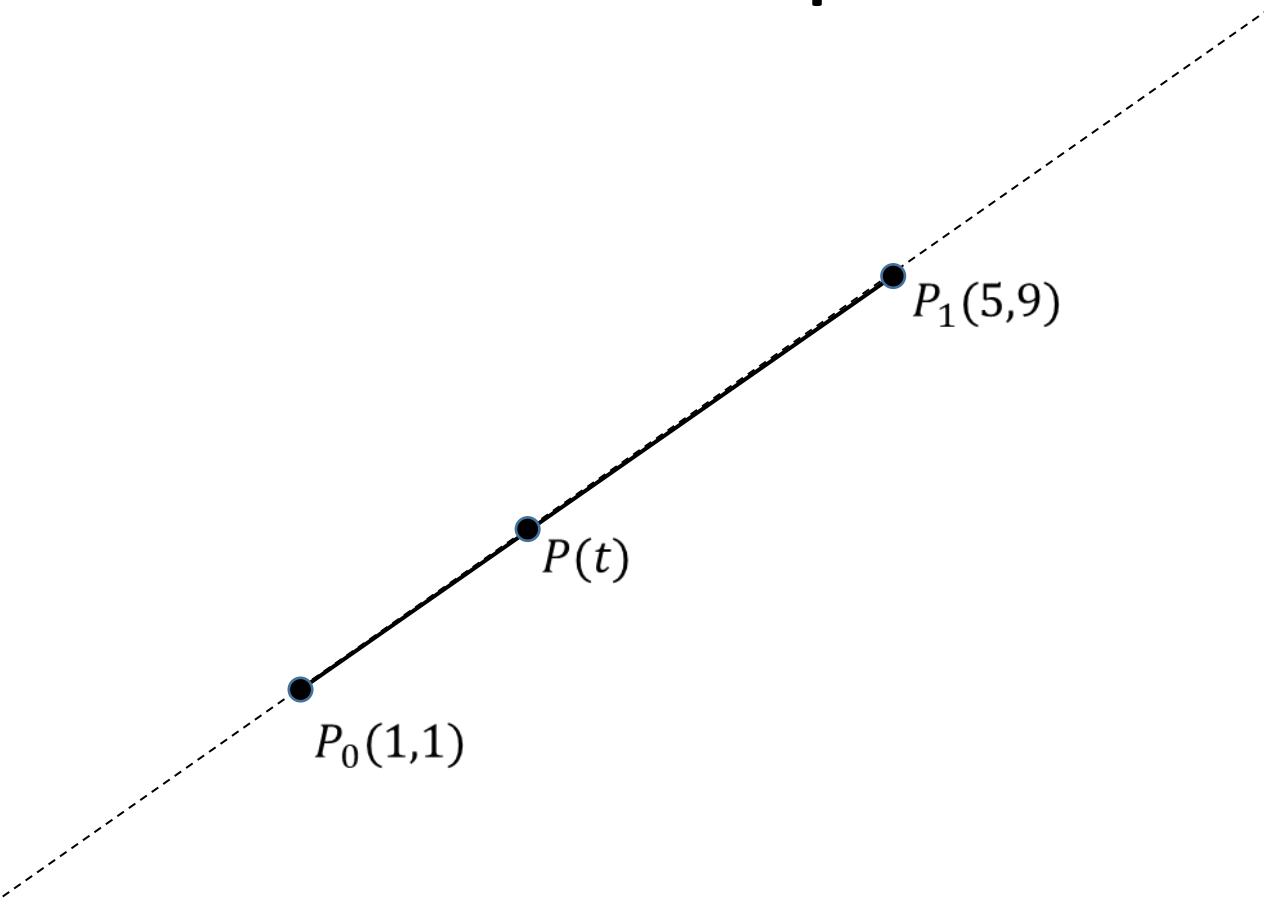


Parametric equation,

$$\begin{aligned} P(t) &= P_0 + t \cdot (P_1 - P_0) \\ &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\ &= (x_0 + t(x_1 - x_0), y_0 + t(y_1 - y_0)) \end{aligned}$$

$$\begin{aligned} P\left(\frac{1}{2}\right) &= (x_0, y_0) + \frac{1}{2} \cdot (x_1 - x_0, y_1 - y_0) \\ &= \left(\frac{x_0 + x_1}{2}, \frac{y_0 + y_1}{2}\right) \end{aligned}$$

Parametric equation of Line



Suppose, endpoints of a line are $(1, 1)$ and $(5, 9)$

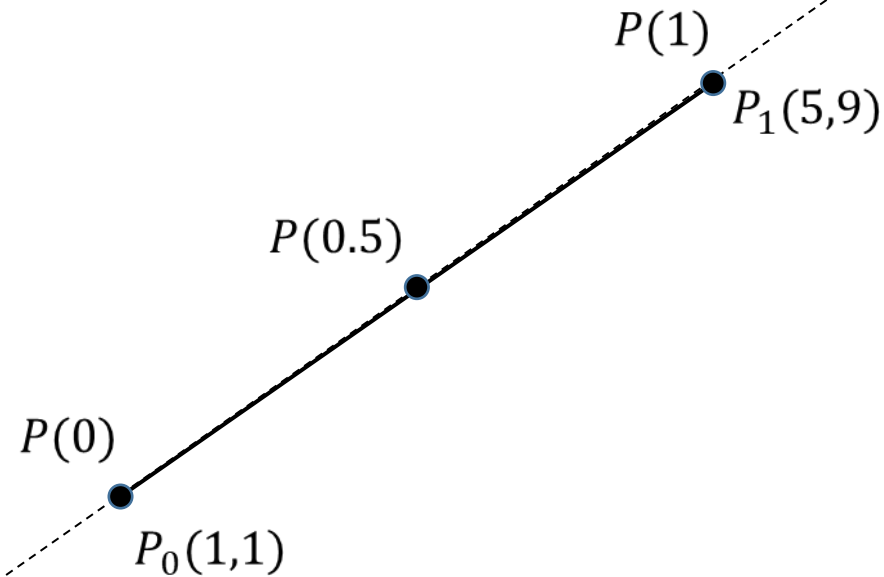
Parametric equation,

$$\begin{aligned} P(t) &= P_0 + t \cdot (P_1 - P_0) \\ &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\ &= (1, 1) + t(5 - 1, 9 - 1) \\ &= (1, 1) + t(4, 8) \\ &= (1 + 4t, 1 + 8t) \end{aligned}$$

$$P(0) = (1, 1)$$

$$P(1) = (5, 9)$$

Parametric equation of Line



Suppose, endpoints of a line are $(1, 1)$ and $(5, 9)$

Parametric equation,

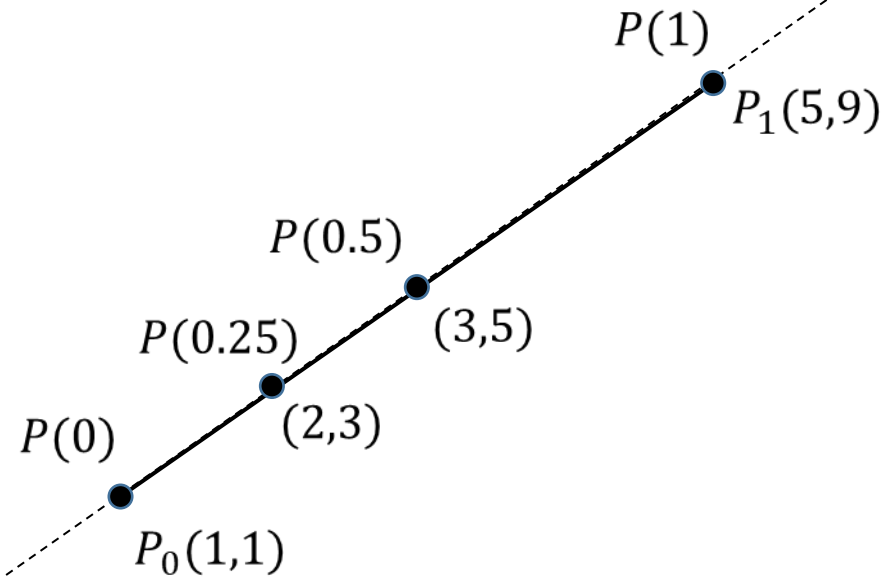
$$\begin{aligned}
 P(t) &= P_0 + t \cdot (P_1 - P_0) \\
 &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\
 &= (1, 1) + t(5 - 1, 9 - 1) \\
 &= (1, 1) + t(4, 8) \\
 &= (1 + 4t, 1 + 8t)
 \end{aligned}$$

$$P(0) = (1, 1)$$

$$P(1) = (5, 9)$$

$$P(0.5) = (1 + 4 \times 0.5, 1 + 8 \times 0.5) = (3, 5)$$

Parametric equation of Line



Suppose, endpoints of a line are (1, 1) and (5, 9)

Parametric equation,

$$\begin{aligned}
 P(t) &= P_0 + t \cdot (P_1 - P_0) \\
 &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\
 &= (1, 1) + t(5 - 1, 9 - 1) \\
 &= (1, 1) + t(4, 8) \\
 &= (1 + 4t, 1 + 8t)
 \end{aligned}$$

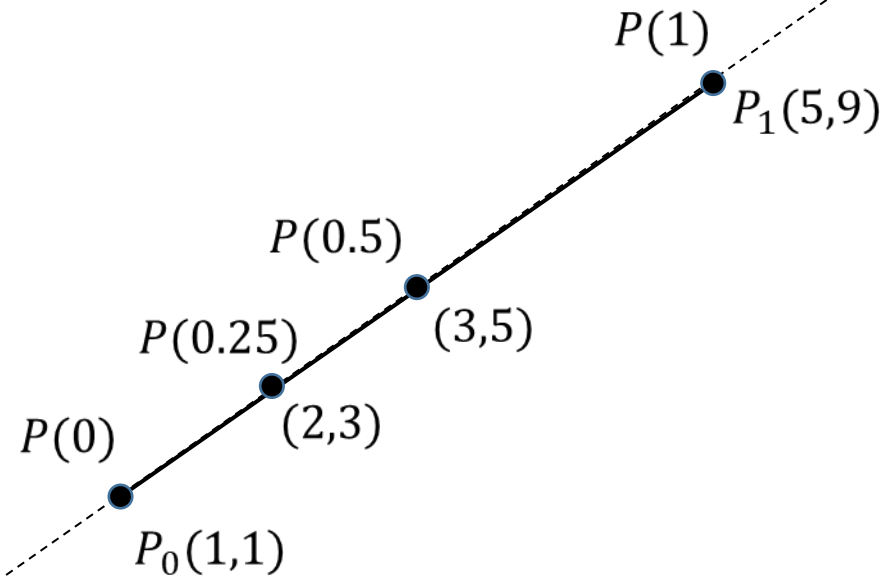
$$P(0) = (1, 1)$$

$$P(1) = (5, 9)$$

$$P(0.5) = (1 + 4 \times 0.5, 1 + 8 \times 0.5) = (3, 5)$$

$$P(0.25) = (1 + 4 \times 0.25, 1 + 8 \times 0.25) = (2, 3)$$

Parametric equation of Line



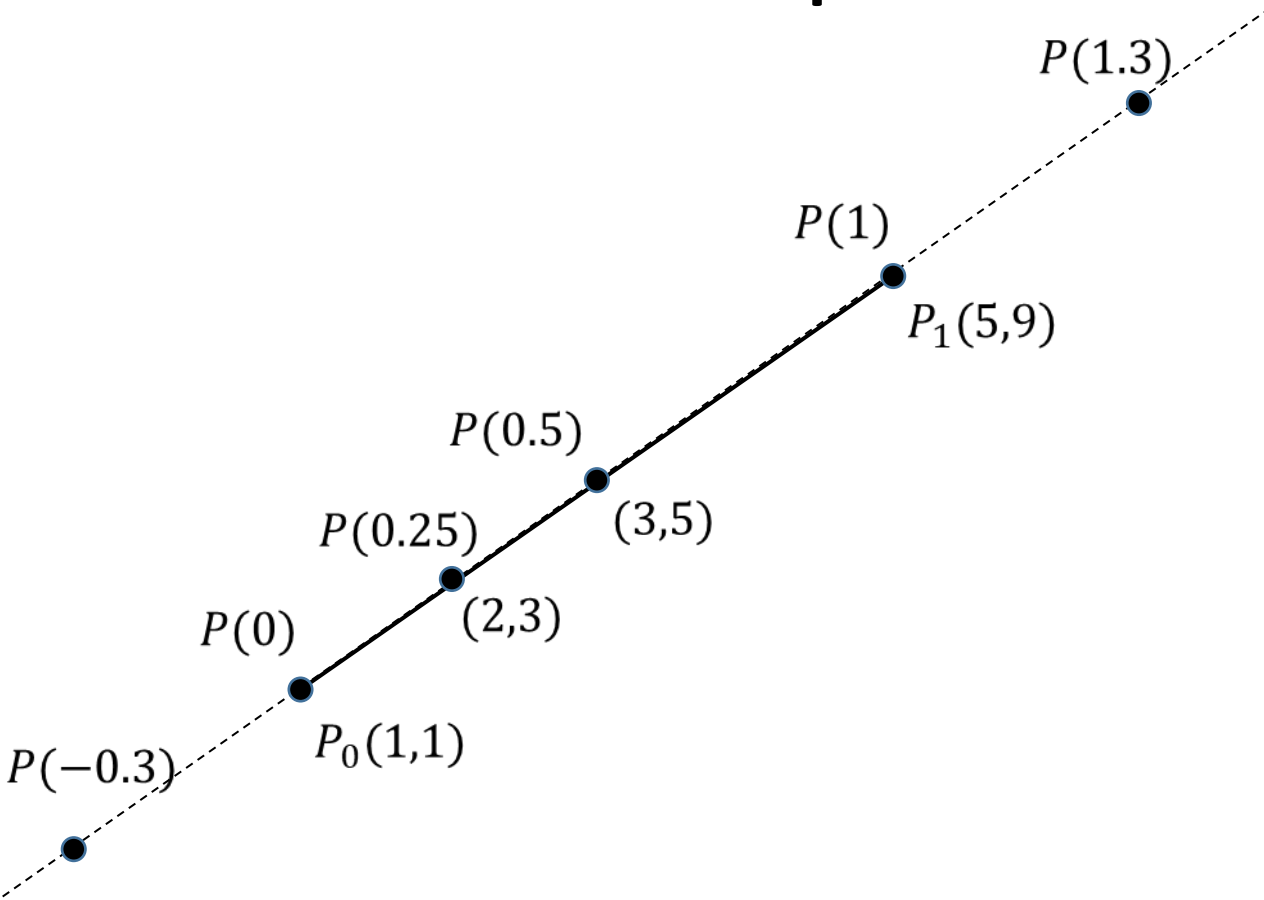
Suppose, endpoints of a line are $(1, 1)$ and $(5, 9)$

Parametric equation,

$$\begin{aligned}
 P(t) &= P_0 + t \cdot (P_1 - P_0) \\
 &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\
 &= (1, 1) + t(5 - 1, 9 - 1) \\
 &= (1, 1) + t(4, 8) \\
 &= (1 + 4t, 1 + 8t)
 \end{aligned}$$

So all points between $0 \leq t \leq 1$ are inside the line segment

Parametric equation of Line

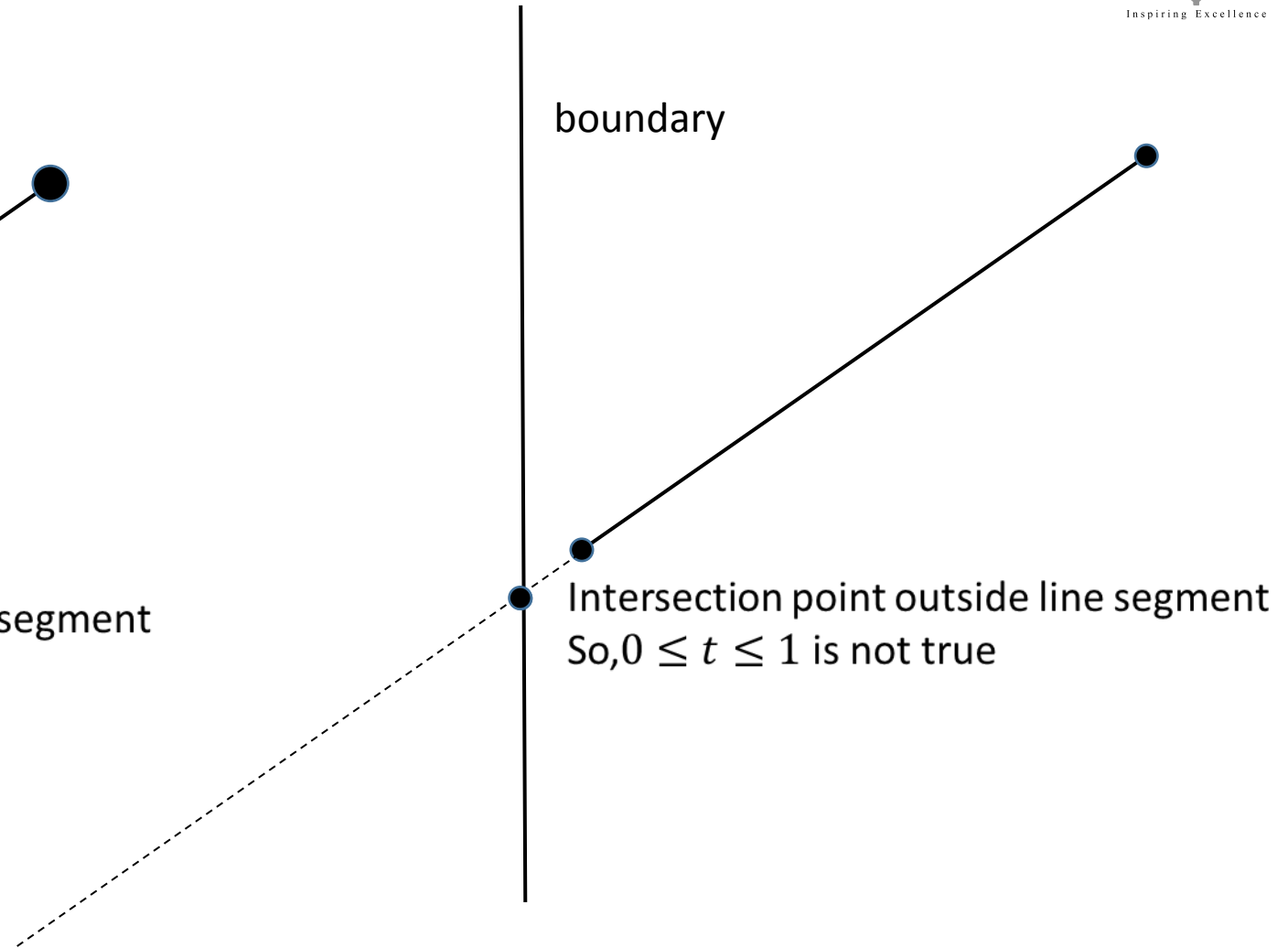
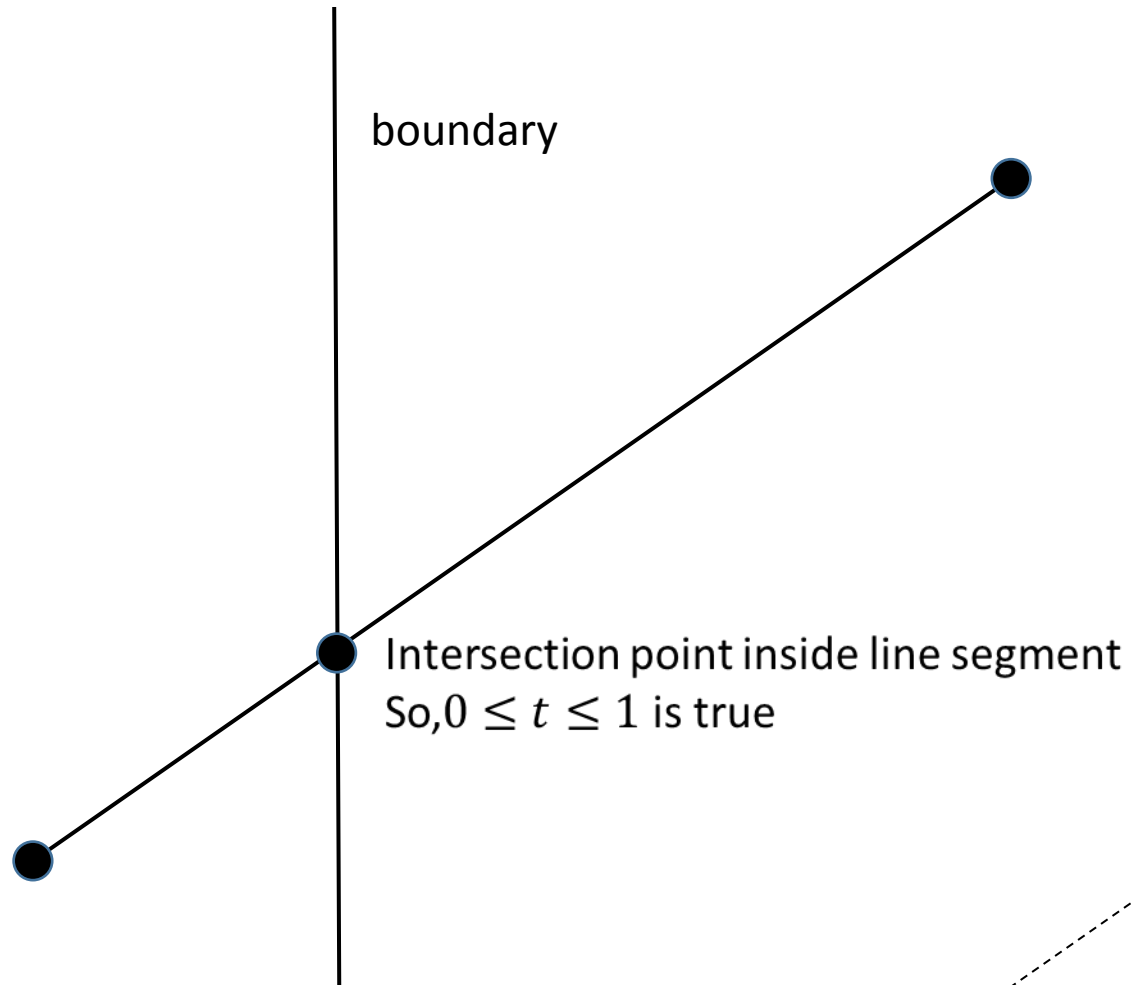


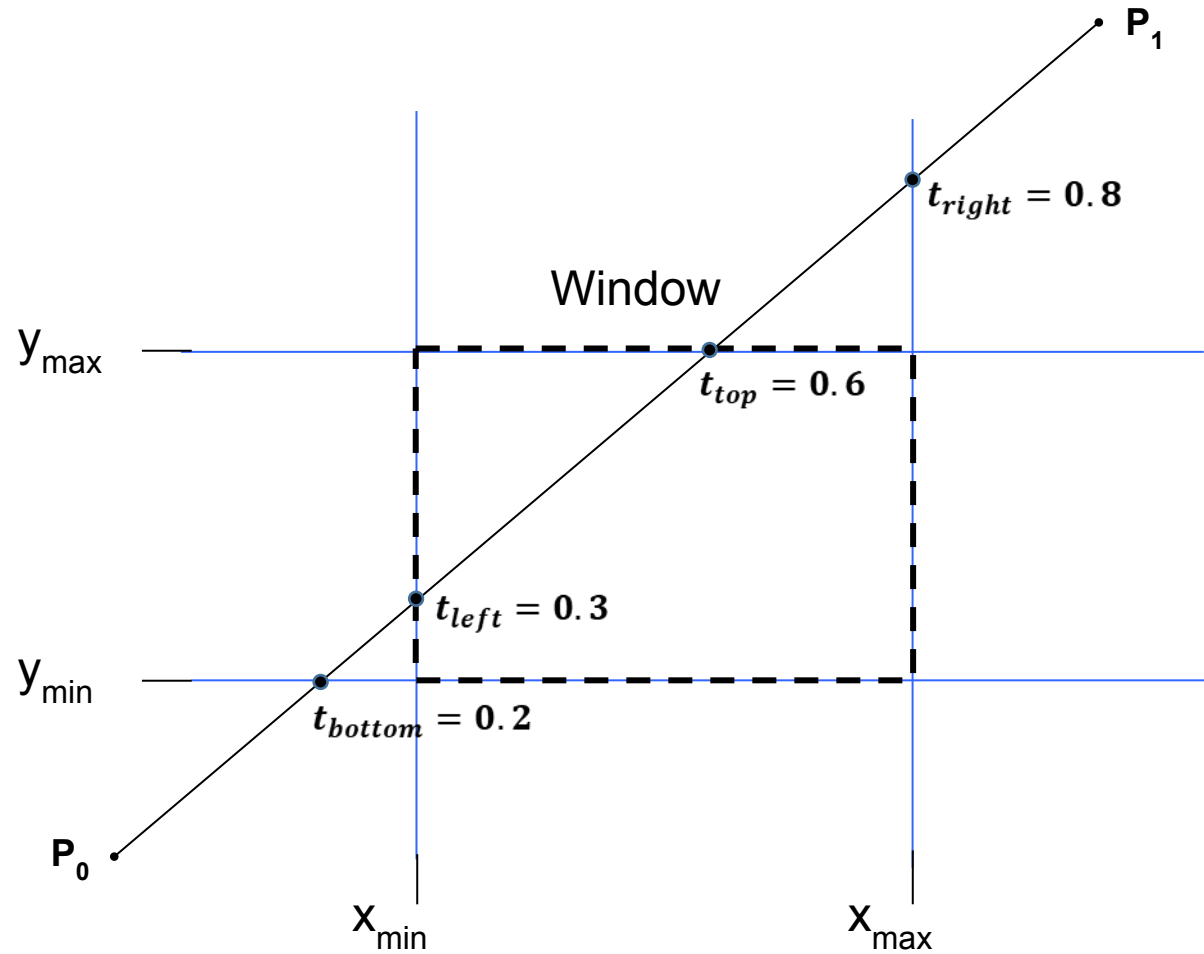
Suppose, endpoints of a line are $(1, 1)$ and $(5, 9)$

Parametric equation,

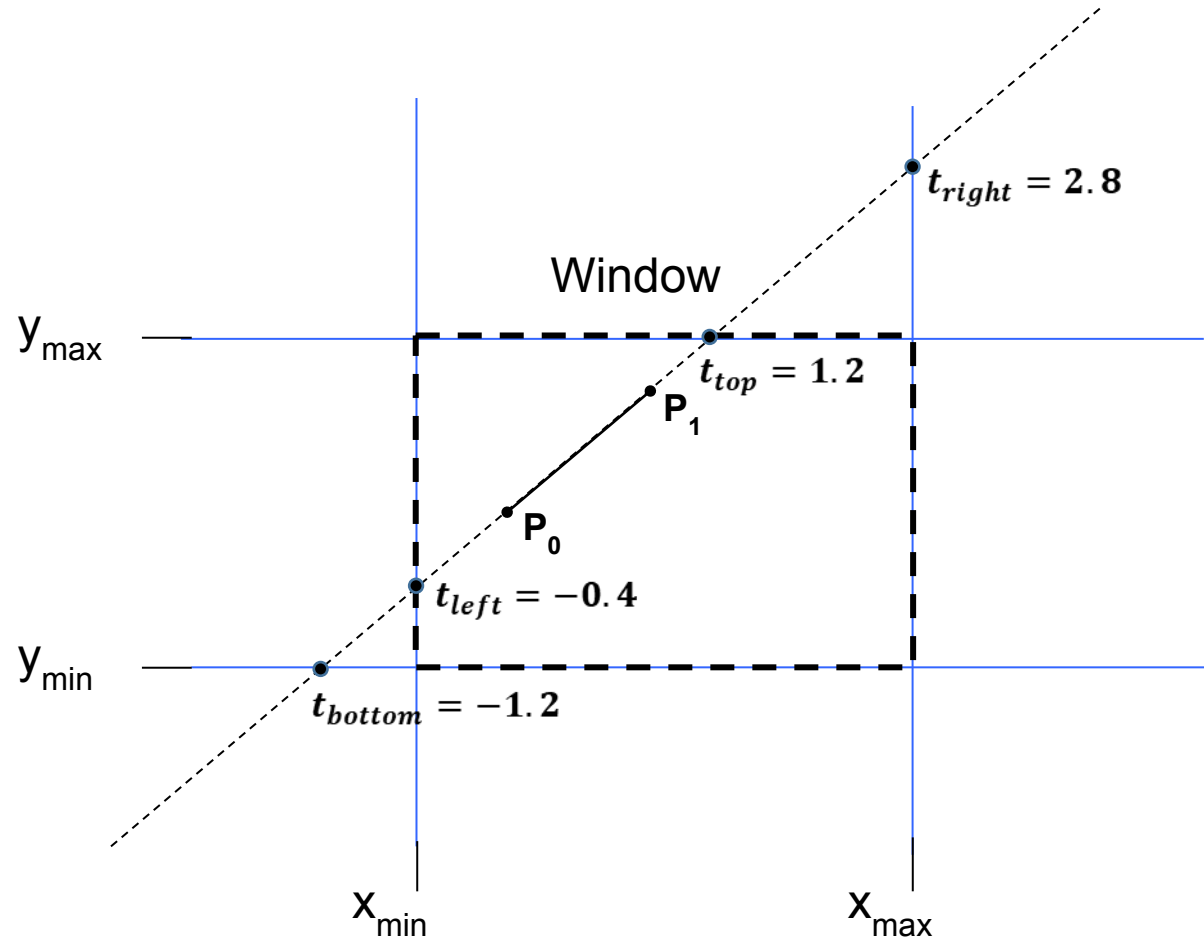
$$\begin{aligned}
 P(t) &= P_0 + t \cdot (P_1 - P_0) \\
 &= (x_0, y_0) + t(x_1 - x_0, y_1 - y_0) \\
 &= (1, 1) + t(5 - 1, 9 - 1) \\
 &= (1, 1) + t(4, 8) \\
 &= (1 + 4t, 1 + 8t)
 \end{aligned}$$

$P(-0.3)$ and $P(1.3)$ are outside the line segment



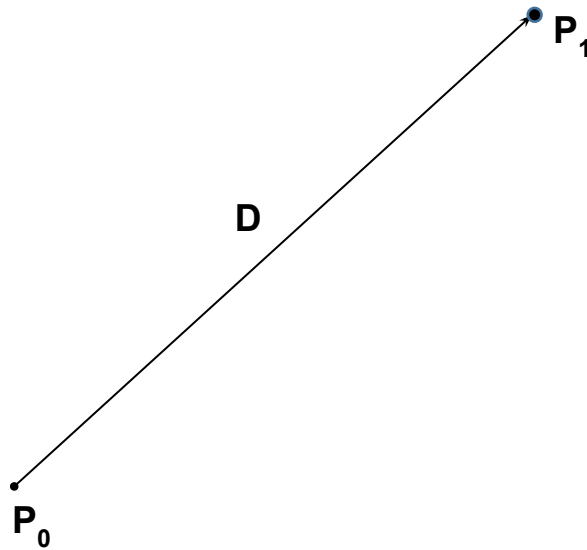


Both endpoints outside
All intersections have t between
0 and 1



Both endpoints inside
All intersections have t outside 0
to 1

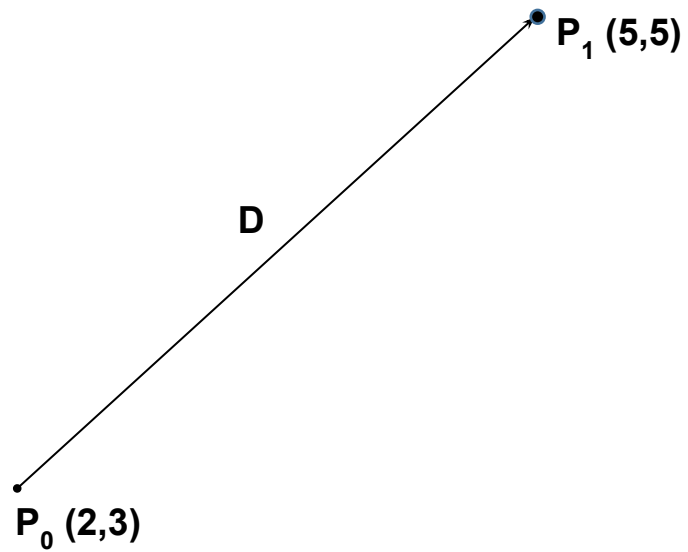
Vector



D is a vector from P_0 to P_1

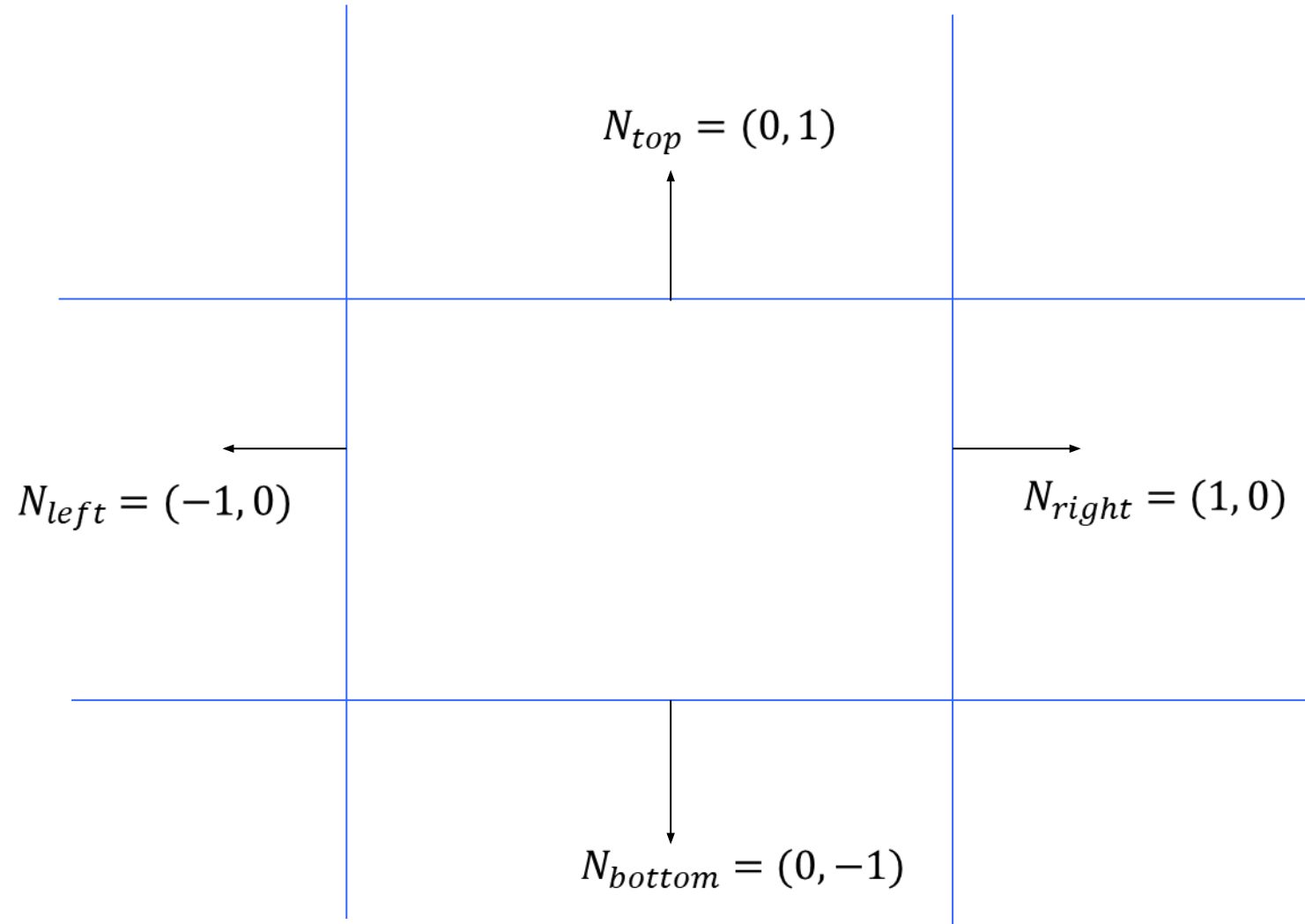
$$D = P_1 - P_0$$
$$= (x_1 - x_0, y_1 - y_0)$$

Vector



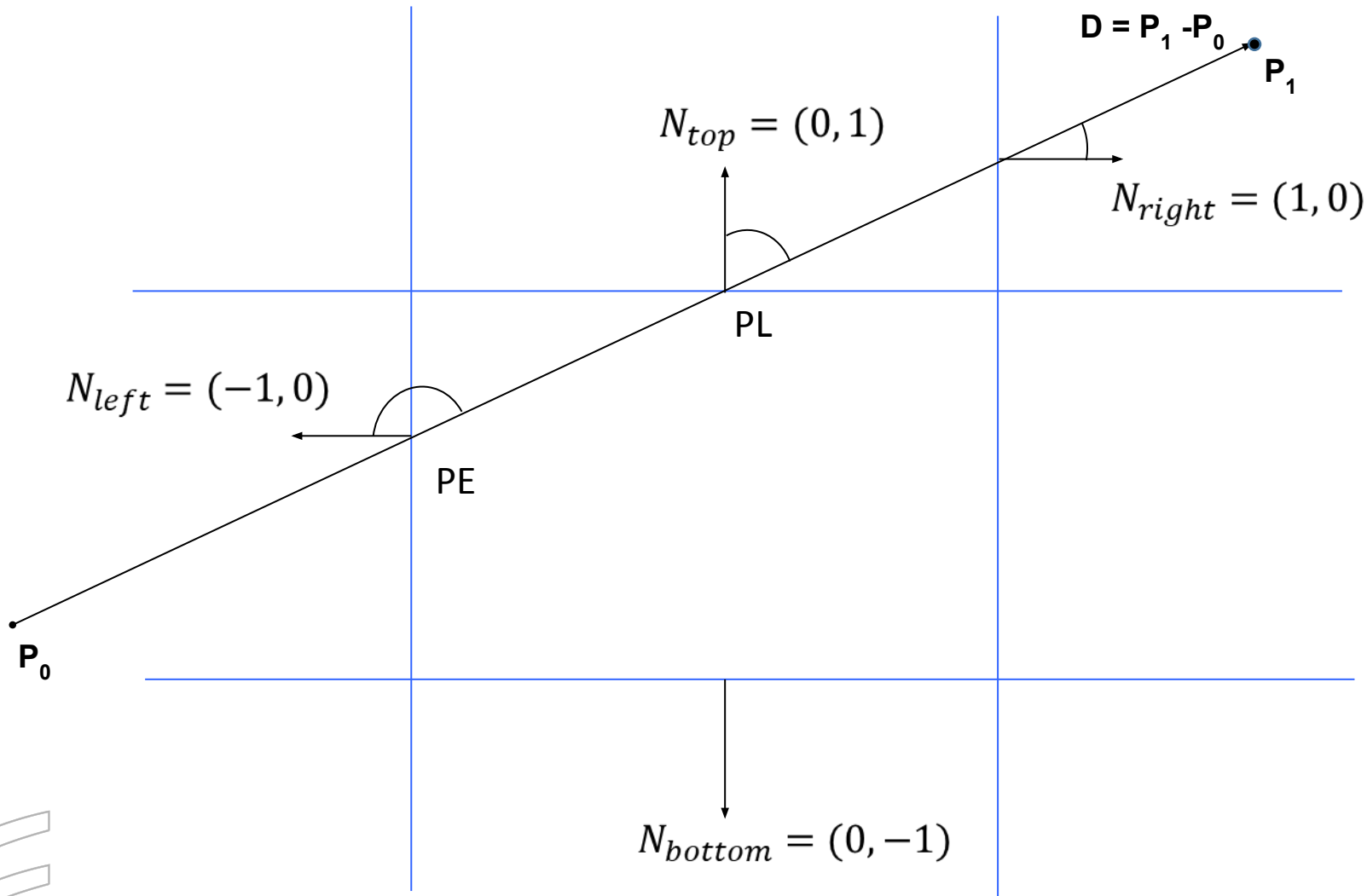
$$\begin{aligned} D &= P_1 - P_0 \\ &= (x_1 - x_0, y_1 - y_0) \\ &= (5 - 2, 5 - 3) \\ &= (3, 2) \\ &\cong 3i + 2j \end{aligned}$$

Normal vectors to boundary



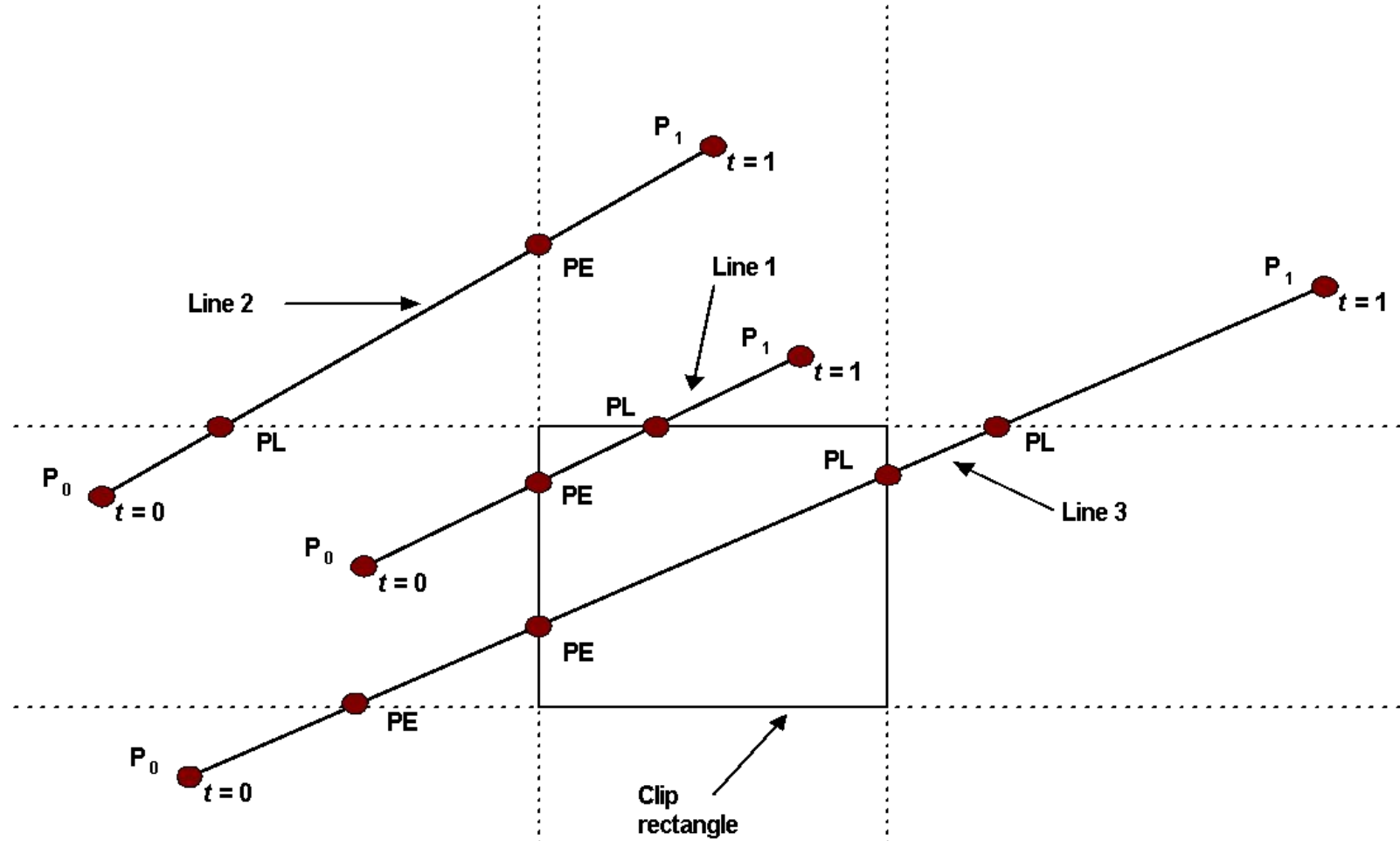
Each N_i is a perpendicular unit vector to each of the boundaries: left, right, top, bottom. N_i points away from the clip window

Normal vectors to boundary



When angle between N_i and D is more than 90 degree-
Or, $N_i \cdot D < 0$
The intersection point is PE
(Potentially entering)

When angle between N_i and D is less than 90 degree-
Or, $N_i \cdot D > 0$
The intersection point is PL
(Potentially leaving)



PE = Potentially Entering

$$N_i \cdot D < 0 \Rightarrow PE \\ \Rightarrow Angle > 90^\circ$$

PL = Potentially Leaving

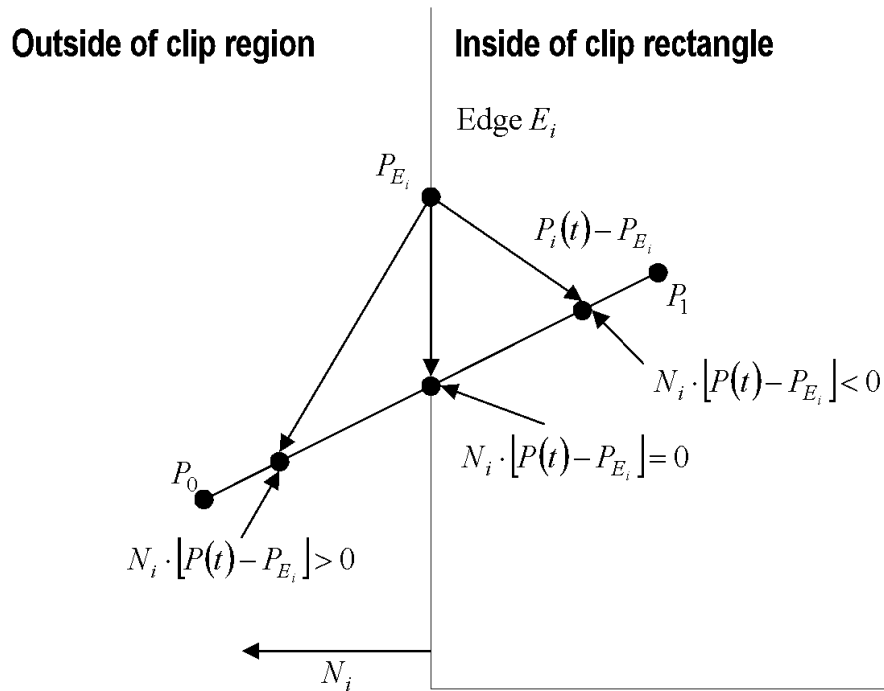
$$N_i \cdot D > 0 \Rightarrow PL \\ \Rightarrow Angle < 90^\circ$$

Cyrus-Beck algo

- Calculate t values of intersection points with each 4 boundaries
- Classify intersection points whether PE/PL
- Select the PE with highest t and the PL with the lowest t
- Using parametric line eqn. find the clipped points



The Cyrus-Beck Algorithm



$$\textit{Line } P_0P_1 : P(t) = P_0 + (P_1 - P_0)t$$

$$N_i \cdot [P(t) - P_{E_i}] = 0$$

$$\Rightarrow N_i \cdot [P_0 + (P_1 - P_0)t - P_{E_i}] = 0$$

$$\Rightarrow N_i \cdot [P_0 + (P_1 - P_0)t - P_{E_i}] = 0$$

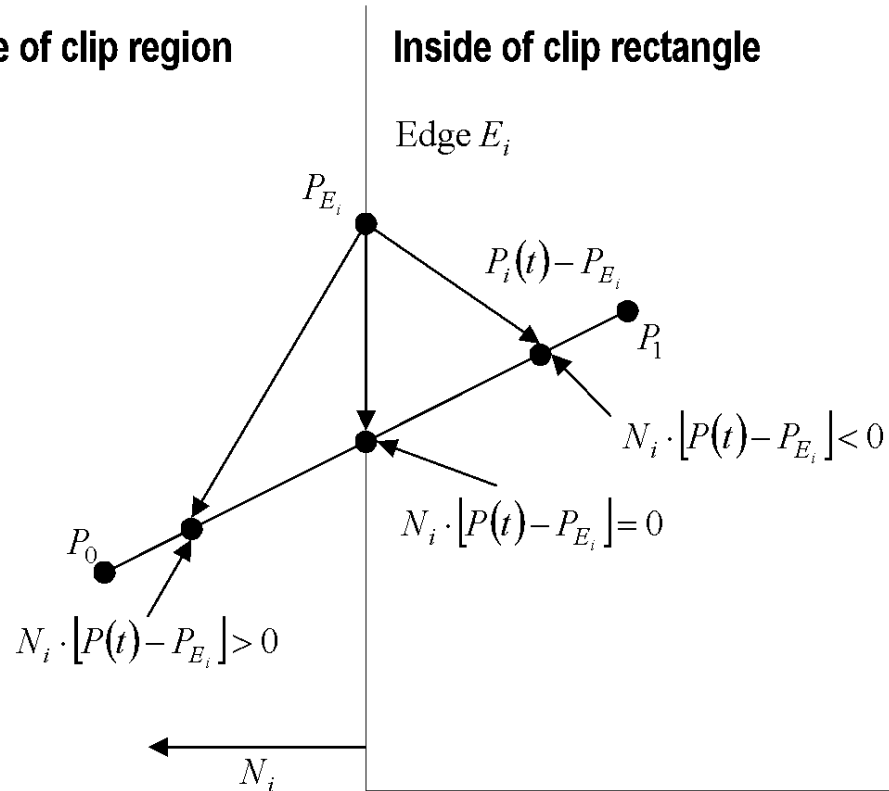
$$\Rightarrow t = \frac{N_i \cdot [P_0 - P_{E_i}]}{-N_i \cdot (P_1 - P_0)}$$

$$\Rightarrow t = \frac{N_i \cdot [P_0 - P_{E_i}]}{-N_i \cdot D}, \quad D = P_1 - P_0$$

The Cyrus-Beck Algorithm

Outside of clip region

Inside of clip rectangle



For left boundary,

$$\begin{aligned}
 t_{left} &= \frac{N_i \cdot [P_0 - P_{E_i}]}{-N_i \cdot [P_1 - P_0]} \\
 &= \frac{(-1, 0) \cdot [(x_0, y_0) - (x_{min}, y)]}{-(-1, 0) \cdot [(x_1, y_1) - (x_0, y_0)]} \\
 &= \frac{(-1, 0) \cdot (x_0 - x_{min}, y_0 - y)}{-(-1, 0) \cdot (x_1 - x_0, y_1 - y_0)} \\
 &= \frac{-1 \times (x_0 - x_{min}) + 0 \times (y_0 - y)}{-(-1 \times (x_1 - x_0) + 0 \times (y_1 - y_0))} \\
 &= \frac{-(x_0 - x_{min})}{(x_1 - x_0)}
 \end{aligned}$$

Similarly,

$$\begin{aligned}
 t_{right} &= \frac{-(x_0 - x_{max})}{(x_1 - x_0)} \\
 t_{top} &= \frac{-(y_0 - y_{max})}{(y_1 - y_0)} \\
 t_{bottom} &= \frac{-(y_0 - y_{min})}{(y_1 - y_0)}
 \end{aligned}$$

The Cyrus-Beck Algorithm

```

Precalculate  $N_i$  and  $P_{Ei}$  for each edge
for (each line segment to be clipped) {
  if ( $P_1 == P_0$ )
    line is degenerated, so clip as a point;
  else {
     $t_E = 0$ ;  $t_L = 1$ ;
    for (each candidate intersection with a clip edge) {
      if ( $N_i \bullet D \neq 0$ ) { /* Ignore edges parallel to line */
        calculate  $t$ ;
        use sign of  $N_i \bullet D$  to categorize as PE or PL;
        if (PE)  $t_E = \max(t_E, t)$ ;
        if (PL)  $t_L = \min(t_L, t)$ ;
      }
    }
    if ( $t_E > t_L$ ) return NULL;
    else return  $P(t_E)$  and  $P(t_L)$  as true clip intersection;
  }
}
  
```


a) Calculate the value of t of the lines given below for all edges and specify whether they are entering or leaving t . [Given $(0,0)$ to $(300,200)$ be the clip region.]

- (i) $(50, -125)$ to $(-100, 225)$.
- (ii) $(-250, 200)$ to $(250, -200)$.
- (iii) $(-250, 200)$ to $(150, 100)$

Also, find the line segment within the clipping window



a)(i) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_0 = 50, y_0 = -125, x_1 = -100, y_1 = 225$$

$$D = (x_1 - x_0, y_1 - y_0) = (-150, 350)$$

Initially, $t_E = 0, t_L = 1$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
Left	(-1,0)	150	$\frac{-(50 - 0)}{-100 - 50} = 0.33$	PL	0	0.33
Right	(1,0)	-150	$\frac{-(50 - 300)}{-100 - 50} = -1.67$	PE	0	0.33
Bottom	(0, -1)	-350	$\frac{-(-125 - 0)}{225 - (-125)} = 0.357$	PE	0.357	0.33
Top	(0, 1)	350	$\frac{-(-125 - 200)}{225 - (-125)} = 0.93$	PL	0.357	0.33

Since $t_E > t_L$, line segment is outside clip window

$$t_{left} = \frac{-(x_0 - x_{min})}{(x_1 - x_0)}$$

$$t_{right} = \frac{-(x_0 - x_{max})}{(x_1 - x_0)}$$

$$t_{top} = \frac{-(y_0 - y_{max})}{(y_1 - y_0)}$$

$$t_{bottom} = \frac{-(y_0 - y_{min})}{(y_1 - y_0)}$$

Algorithm

Precalculate N_i and P_{Ei} for each edge

for (each line segment to be clipped) {

if ($P_1 == P_0$)

line is degenerated, so clip as a point;

else {

$t_E = 0; t_L = 1;$

for (each candidate intersection with a clip edge) {

if ($N_i \cdot D \neq 0$) { /* Ignore edges parallel to line */
calculate t;

use sign of $N_i \cdot D$ to categorize as PE or PL;

if (PE) $t_E = \max(t_E, t);$

if (PL) $t_L = \min(t_L, t);$

}

}

if ($t_E > t_L$) return NULL;

else return $P(t_E)$ and $P(t_L)$ as true clip intersection;

}

a)(ii) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_0 = -250, y_0 = 200, x_1 = 250, y_1 = -200$$

$$D = (x_1 - x_0, y_1 - y_0) = (500, -400)$$

$$\text{Initially, } t_E = 0, t_L = 1$$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
Left	(-1,0)	-500	$\frac{-(-250 - 0)}{500}$ = 0.5	PE	0.5	1
Right	(1,0)	500	$\frac{-(-250 - 300)}{500}$ = 1.1	PL	0.5	1
Bottom	(0, -1)	400	$\frac{-(200 - 0)}{-400}$ = 0.5	PL	0.5	0.5
Top	(0, 1)	-400	$\frac{-(200 - 200)}{-400}$ = 0	PE	0.5	0.5

P(0.5) and P(0.5) (same point) are the true clip intersection

$$\begin{aligned}
 P(0.5) &= (x_0, y_0) + 0.5 \times D \\
 &= (-250, 200) + 0.5 \times (500, -400) \\
 &= (0, 0)
 \end{aligned}$$

$$\begin{aligned}
 t_{left} &= \frac{-(x_0 - x_{min})}{(x_1 - x_0)} \\
 t_{right} &= \frac{-(x_0 - x_{max})}{(x_1 - x_0)} \\
 t_{top} &= \frac{-(y_0 - y_{max})}{(y_1 - y_0)} \\
 t_{bottom} &= \frac{-(y_0 - y_{min})}{(y_1 - y_0)}
 \end{aligned}$$

Algorithm

Precalculate N_i and P_{Ei} for each edge

for (each line segment to be clipped) {

if ($P_1 == P_0$)

line is degenerated, so clip as a point;

else {

$t_E = 0; t_L = 1;$

for (each candidate intersection with a clip edge) {

if ($N_i \cdot D \neq 0$) { /* Ignore edges parallel to line */

calculate t;

use sign of $N_i \cdot D$ to categorize as PE or PL;

if (PE) $t_E = \max(t_E, t);$

if (PL) $t_L = \min(t_L, t);$

}

}

if ($t_E > t_L$) return NULL;

else return $P(t_E)$ and $P(t_L)$ as true clip intersection;

}

}

a)(iii) boundary:

$$x_{min} = 0, x_{max} = 300, y_{min} = 0, y_{max} = 200$$

points:

$$x_0 = -250, y_0 = 200, x_1 = 150, y_1 = 100$$

$$D = (x_1 - x_0, y_1 - y_0) = (400, -100)$$

$$\text{Initially, } t_E = 0, t_L = 1$$

Boundary	N_i	$N_i \cdot D$	t	PE/PL	t_E	t_L
Left	(-1,0)	-400	$\frac{-(-250 - 0)}{400}$ $= 0.625$	PE	0.625	1
Right	(1,0)	400	$\frac{-(-250 - 300)}{400}$ $= 1.375$	PL	0.625	1
Bottom	(0, -1)	100	$\frac{-(200 - 0)}{-100}$ $= 2$	PL	0.625	1
Top	(0, 1)	-100	$\frac{-(200 - 200)}{-100}$ $= 0$	PE	0.625	1

P(0.625) and P(1) are the true clip intersection

$$\begin{aligned} P(0.625) &= (x_0, y_0) + 0.625 \times D \\ &= (-250, 200) + 0.625 \times (400, -100) \\ &= (0, 137.5) \end{aligned}$$

$$P(1) = (150, 100)$$

(0, 137.5) and (150, 100) are the endpoints of the clipped line

$$\begin{aligned} t_{left} &= \frac{-(x_0 - x_{min})}{(x_1 - x_0)} \\ t_{right} &= \frac{-(x_0 - x_{max})}{(x_1 - x_0)} \\ t_{top} &= \frac{-(y_0 - y_{max})}{(y_1 - y_0)} \\ t_{bottom} &= \frac{-(y_0 - y_{min})}{(y_1 - y_0)} \end{aligned}$$

Algorithm

Precalculate N_i and P_{Ei} for each edge

for (each line segment to be clipped) {

if ($P_1 == P_0$)

line is degenerated, so clip as a point;

else {

$t_E = 0$; $t_L = 1$;

for (each candidate intersection with a clip edge) {

if ($N_i \cdot D \neq 0$) { /* Ignore edges parallel to line */

calculate t;

use sign of $N_i \cdot D$ to categorize as PE or PL;

if (PE) $t_E = \max(t_E, t)$;

if (PL) $t_L = \min(t_L, t)$;

}

}

if ($t_E > t_L$) return NULL;

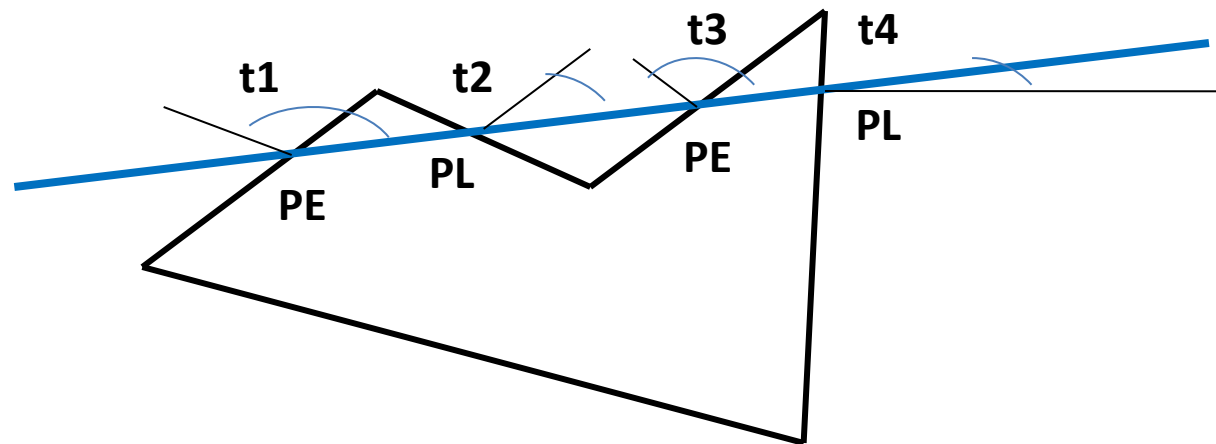
else return $P(t_E)$ and $P(t_L)$ as true clip intersection;

}

}

Cyrus-Beck Parametric Line Clipping Algorithm

- Advantage
 - Works with polygons too (not only with clip rectangles)
 - Works in 3D scenario (polyhedrons)
- Problem
 - Does not work with **concave** polygon clip region



$$\begin{aligned} t_E &= t_3 \\ t_L &= t_2 \\ t_E &> t_L \end{aligned}$$

So the whole line is discarded though some segments should be displayed

Simulation Exercise

1. Given $(-30, -30)$ and $(30, 30)$ are the corners of the clip rectangle. The endpoints of a line are given as $(35, 37)$ and $(-38, -20)$. Using the Cohen-Sutherland line clipping algorithm, identify whether the lines are Accepted, Rejected, or Partial (partially accepted) only from the outcodes. Does the Cyrus-Beck line clipping algorithm produce the same output?
2. Given $(0, 0)$ and $(400, 300)$ are the corners of the clip rectangle. The endpoints of a line are given as $(-50, 350)$ and $(450, 330)$. Using the Cohen-Sutherland line clipping algorithm, identify whether the lines are Accepted, Rejected, or Partial (partially accepted) only from the outcodes. Does the Cyrus-Beck line clipping algorithm produce the same output?
3. The top-right corner of a clipping rectangle is $(20, 60)$. The window is 60 units wide and 40 units tall. The endpoints of a line segment are $(7, 16)$ and $(85, 75)$. Apply the Cohen-Sutherland Algorithm, and identify whether the line is "partially inside", "completely inside", or "completely outside". If it is the first case, run the algorithm to calculate new endpoints for the line segment so that it is inside the clipping window. Does the Cyrus-Beck line clipping algorithm produce the same output?

Simulation Exercise

4. A clipping window is 70 units wide and 30 units tall, and has its center at (0, 35). The endpoints of a line segment are (-45, 65) and (23, -10). Apply the Cohen-Sutherland Algorithm, and identify whether the line is "partially inside", "completely inside", or "completely outside". If it is the first case, run the algorithm to calculate new endpoints for the line segment so that it is inside the clipping window. Does the Cyrus-Beck line clipping algorithm produce the same output?
5. Given a line segment from (10, 60) to (25, 30). Construct the parametric equation $P(t)$ of the line. Using the parametric equation, determine the coordinates of the point where $t = 3/5$.

Reference:

Computer Graphics: Principles and Practice: John F. Hughes, James D. Foley, Andries van Dam, Steven K. Feiner (2nd Edition)

Chapter: 3.12, 3.14

